Agent-GhibliLock: Deterministic Wavelet-Locked Cross-Modal Style Transfer and ISO/IEC 19566-5 C2PA-Inspired Provenance Compilation for Multimodal Generative Pipelines

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***Abstract*—Enterprise multimodal workflows increasingly chain vision-language models (VLMs) with image-conditioned or adapter-guided latent diffusion image and video synthesizers. When an upstream VLM translates a source photograph into natural-language style instructions and a downstream diffusion model synthesizes restyled pixels (Pixels $\rightarrow$ Prose $\rightarrow$ Pixels), three coupled failure modes arise, which we term *The Captioning Conundrum*: (1) *high-frequency structural phase drift*, where even under macro-registered image or adapter conditioning, fine-grained corporate emblems, typographic glyphs, and perspective rails are degraded across the low-entropy text and $8\times$ latent VAE bottlenecks; (2) *temporal luminance flicker in video Groups of Pictures (GOPs)*, where frame-wise stochastic denoising induces severe exposure variance while naive temporal smoothing blurs moving edges; and (3) *cryptographic provenance severance and host-signal interference*, where container-level Coalition for Content Provenance and Authenticity (C2PA) manifests are stripped during cross-modal regeneration and social-platform re-encoding while blind additive spread-spectrum watermarks suffer host-signal interference. We present Agent-GhibliLock, a deterministic, CPU-executable post-processing compiler that reconciles high-frequency structural preservation, target palette alignment, temporal GOP stability, and dual-layer provenance attestation. First, Agent-GhibliLock decomposes CIELAB luminance (L^*) via an orthonormal K-level 2D Discrete Haar Wavelet Transform (DWT_2), supporting both *global* subband locking and *spatially masked region-of-interest (ROI)* detail locking (weighted by 5×5 Sobel directional coherence so macro-registered foreground emblems and rails are phase-locked while non-ROI background regions retain smooth painterly gouache brushwork), and proves that arbitrary spatial detail masks $M_k(u,v)\in[0,1]$ preserve exact spatial mean luminance $\mu(\widehat{L}^*)=\mu(\widetilde{L}^*_{\text{cand}})$ (Theorem 1). Second, it aligns full 3D (L^*,a^*,b^*) color distributions via Iterative Sliced-Wasserstein Optimal Transport (SWOT) over $D=3M+3=27$ orthonormal axes ($I_3\cup\{Q_m\}_{m=1}^{M}\subset O(3)$) with deterministic lexicographical tie-breaking, explicit post-Step-A luminance mean re-anchoring, and a decoupled $(\tau_\gamma,\tau_\alpha)$ closed-loop controller satisfying an exact deterministic root-mean-square (RMS) Wasserstein perturbation bound (Theorem 2). Third, for GOP video synthesis, it suppresses inter-frame flicker via translation-compensated, innovation-gated wavelet LL_K exponential moving average (EMA) filtering while committing frame hashes to a binary Merkle tree root. Fourth, it binds hard and soft provenance by injecting a compact C2PA-inspired ISO/IEC 19566-5 JUMBF binary manifest box (`caBX` PNG chunk with IEEE 802.3 CRC-32 and keyed HMAC-SHA256) alongside a blind Quantization-Index-Modulation (QIM) watermark in mid-band 8×8 2D-DCT AC luminance coefficients supporting both $M_{\text{coord}}=4$ ($R=\lfloor B/16\rfloor$) and $M_{\text{coord}}=5$ odd-repetition ($R=5B/64$) clipped-soft/hard majority voting, keyed dither K_{secret}, analytical AC ℓ_1 rounding bounds ($<\Delta_{\text{QIM}}/4$), and blind coset tamper localization (Theorem 3). Across a stratified $N=600$ benchmark (evaluating 9 paradigms including a ControlNet $8\times$-VAE adapter baseline, $K\in\{1,2,3\}$ wavelet depth ablations, and disaggregated 4×25 adversarial provenance regimes), live multi-model Vertex AI Gemini evaluations (`gemini-2.5-flash`, `gemini-2.5-pro`, `gemini-3-flash-preview`), and real natural photographic evaluations (`grace_hopper.jpg` at 64^2–512^2 and `Minduka_Present_Blue_Pack.png`), Agent-GhibliLock achieves $\mathrm{SSIM}_{\text{edge}}=0.9967$ (0.9977 ROI-masked; 0.9929–0.9966 on real photographs), non-circular $\mathrm{SSIM}_{\text{full}}=0.9752$, Sobel gradient correlation $\rho_{\text{grad}}=0.8943$, post-export $\widehat{W}_2^{\text{out}}=1.9614$ (held-out $\widehat{SW}_2^{\text{eval}}=2.0741$), 99.9% GOP flicker reduction, and 100.0% tamper detection and blind watermark bit recovery under container stripping, ±2 LSB noise, and lossy JPEG $Q=85$ recompression.**

***Index Terms*—Cross-Modal Style Transfer, Discrete Haar Wavelet Transform, Sliced-Wasserstein Optimal Transport, C2PA JUMBF Provenance, Quantization Index Modulation, 2D-DCT Watermarking, Multimodal Agentic Pipelines.**

I. Introduction

Modern enterprise creative and media workflows frequently orchestrate multi-stage generative pipelines in which a vision-language model (VLM) inspects an input asset I_{src} and emits structured scene descriptions, stylistic prompts, and conditioning priors for a downstream image-conditioned or adapter-guided latent diffusion synthesizer [1]–[4]. Unlike unconstrained text-to-image generation from scratch (where camera pose and macro-layout are synthesized freely), enterprise restyling pipelines preserve coarse scene layout via image-to-image initialization [3] or spatial adapters [4] while relying on VLM-generated prose (Pixels $\rightarrow$ Prose $\rightarrow$ Pixels) to drive artistic palette and brushwork translation (e.g., converting live-action photography into hand-painted Studio Ghibli cel-shaded gouache aesthetics). Even under macro-spatial conditioning, routing visual semantics through a discrete token bottleneck and a downsampled latent autoencoder introduces

three coupled failure modes that we term **The Captioning Conundrum**:

1) **High-Frequency Structural Phase Drift:** Natural language tokens have low spatial entropy (10^2–10^3 tokens $\approx$ 1–2 KB) compared to the high-frequency phase spectrum of an image ($H \times W \times 3$ bytes). Even when paired with iterative VLM-as-a-Judge retries, image-conditioned SDEdit [3], or latent $8\times$-downsampled ControlNet edge/tile adapters [4], fine-scale corporate logos (such as a 2 px geometric diamond shirt emblem), typographic glyphs, eyeglasses, collar insignia, and diagonal architectural rails suffer subband phase drift, spatial blurring across the $8\times$ VAE bottleneck, or local stroke hallucination [5].
2) **Temporal Luminance Flicker in Video GOPs:** When frame-by-frame or chunked diffusion synthesizers generate stylized video Groups of Pictures (GOPs), independent stochastic denoising trajectories induce severe frame-to-frame low-frequency luminance and exposure jumps ($\sigma^2_{\text{flicker}} > 340$). While naive pixel-space temporal exponential moving average (EMA) filtering blurs moving foreground edges, classical blind video temporal consistency methods such as Bonneel *et al.* [6] suppress flicker by solving dense optical-flow-warped gradient-domain screened Poisson equations at significant per-frame computational cost.
3) **Cryptographic Provenance Severance and Host-Signal Interference:** Regenerating pixels through an upstream diffusion decoder severs container-level Coalition for Content Provenance and Authenticity (C2PA) manifests [7]–[9], and social media platforms routinely strip ancillary metadata chunks (`caBX`, EXIF, XMP) upon upload. In the pixel domain, classical blind additive spread-spectrum DCT watermarks [20] suffer from severe Host-Signal Interference (HSI) when decoded without the unwatermarked host image. While modern deep neural watermarking frameworks such as TrustMark [10] and SynthID-Image [19] overcome HSI and geometric distortions via learned encoder-decoder networks, they require GPU neural inference and do not provide analytical sub-Gaussian bit-error bounds or zero-dependency 8×8 block-level algebraic coset tamper localization, motivating host-rejecting lattice Quantization Index Modulation (QIM) [21].

Existing neural style transfer methods [11] and wavelet whitening-and-coloring transforms (WCT^2) [12] preserve spatial contours by operating directly on encoder-decoder feature maps, but rely on second-order Gaussian moment matching that fails to capture multi-modal cel-shaded gouache color distributions and provides neither closed-loop Wasserstein guarantees nor cryptographic C2PA/watermark provenance. Conversely, classical 1D/3D optimal transport (OT) color grading [16]–[18] aligns empirical color histograms, but applying raw pointwise rank-sorting to unconstrained diffusion candidates introduces contour staircasing and destroys source high-frequency edge fidelity unless paired with post-transport gradient-domain Poisson reconstruction [16] or multi-scale regularization [17], [18].

To bridge this gap in a single deterministic pass without post-hoc Poisson PDE solvers or GPU watermark networks, we introduce **Agent-GhibliLock**, a zero-GPU-dependency cross-modal compilation engine that operates as a verifiable post-processing filterbank and provenance compiler on macro-registered candidate assets. To ensure complete experimental clarity regarding live API execution versus offline benchmark reproducibility, our evaluation explicitly separates: (i) **live Stage-0 VLM visual-to-text decomposition calls** executed against production Google Cloud Vertex AI endpoints (`gemini-2.5-flash`, `gemini-2.5-pro`, and `gemini-3-flash-preview` in `project="<gcp-project-id>"`, `location="global"`), where the VLM's emitted `structural_retention_prior` directly parameterizes candidate generation, (ii) a **controlled parametric candidate-degradation harness** over $N = 600$ stratified trials that isolates deterministic compiler convergence from stochastic diffusion seed variance, and (iii) a **Real Natural Photographic Evaluation** across multi-resolution portraits and product photography (`grace_hopper.jpg` at 64×64 to 512×512 and `Minduka_Present_Blue_Pack.png` at 128×128).

A. Key Contributions

1) **Global and Spatially Masked ROI Haar Wavelet Detail Locking with Exact Mean Conservation (Theorem 1):** We formulate an orthonormal K-level 2D Discrete Haar Wavelet Transform ($\text{DWT}_2/\text{IDWT}_2$) [13]–[15] on CIELAB luminance L^* that supports both global subband injection and spatially masked ROI detail locking ($\Gamma_{k,\tau}(u,v) = \gamma_\tau M_k(u,v)$ derived via 5×5 Sobel directional coherence and 2×2 block max-pooling). We prove algebraically that multiplying Haar detail subbands (LH_k, HL_k, HH_k) by any spatially varying mask $M_k(u,v) \in [0,1]$ preserves the exact spatial mean luminance $\mu(\widehat{L}^*) = \mu(\widetilde{L}^*_{\text{cand}})$ because every 2×2 Haar detail synthesis basis vector has zero spatial sum.
2) **3D Sliced-Wasserstein Optimal Transport with Exact Orthonormal-Frame Perturbation Bound (Theorem 2):** We align 3D CIELAB (L^*, a^*, b^*) distributions via iterative 1D optimal transport over $D = 3M + 3 = 27$ orthonormal axes ($Q_0 = I_3$ plus $M = 8$ Haar-uniform $O(3)$ frames via positive-diagonal QR factorization) with stable lexicographical tie-breaking (`mergesort`), explicit post-Step-A luminance mean re-anchoring, and a decoupled $(\tau_\gamma, \tau_\alpha)$ closed-loop controller. Denoting by $\mu_{\widetilde{I}_{\text{cand}}}$, $\mu_{\widehat{I}}$, and $\mu_{Y_{\text{ref}}}$ the empirical 3D CIELAB pixel distributions of the Step-A color-transported candidate $\widetilde{I}_{\text{cand}}$, the Step-B wavelet-locked image $\widehat{I}$, and the target style palette Y_{ref}, respectively, and by $\|\cdot\|_F$ the Frobenius norm over the $H \times W$ luminance plane, we prove via the or-

thogonality identity $\sum_{j=1}^{3} Q_{m,1j}^2 = 1$ ($\forall Q_m \in O(3)$) an exact deterministic bound on the root-mean-square (RMS) Sliced-Wasserstein perturbation $|\widehat{W}_2(\mu_{\widehat{I}}, \mu_{Y_{\text{ref}}}) - \widehat{W}_2(\mu_{\widetilde{I}_{\text{cand}}}, \mu_{Y_{\text{ref}}})| \le (3HW)^{-1/2} \|\widehat{L}^* - \widetilde{L}^*_{\text{cand}}\|_F$ with zero Monte Carlo slack.

3) **Motion-Compensated and Motion-Gated Video GOP Wavelet EMA Deflickering:** For multi-frame video synthesis, we formulate a translation-compensated ($\mathcal{T}_{\Delta u_t, \Delta v_t} \overline{LL}_K^{(t-1)}$), innovation-gated temporal EMA filter on the coarse LL_K wavelet subband that eliminates global pre-clamp luminance flicker ($\sigma^2_{\text{flicker,pre-clamp}} < 10^{-12}$, post-sRGB $\sigma^2_{\text{flicker}} = 0.0003$) and prevents motion ghosting on fast-moving foreground objects while binding frame digests into a C2PA-inspired GOP Merkle root.

4) **Dual-Layer C2PA-Inspired JUMBF (`caBX`) Container and Blind Mid-Band 2D-DCT QIM Watermark with Exact AC Rounding Bounds (Theorem 3):** We combine a compact ISO/IEC 19566-5 JUMBF `caBX` PNG chunk with a blind mid-band 8×8 2D-DCT AC QIM watermark supporting $M_{\text{coord}} \in \{4, 5\}$ ($R = 5$ odd repetition voting at 64×64, $\gcd(5, 64) = 1$), optional keyed dither K_{secret}, analytical AC ℓ_1 rounding bounds ($\max_{\mathcal{M}_4} |\eta_b^{\text{export}}| \le 3.3486 < 3.5 = \Delta_{\text{QIM}}/4$), and clipped-soft/hard majority-voted blind coset tamper localization requiring zero source registry.

II. System Architecture and Mathematical Formulation

A. Stage 0: Cross-Modal Visual-to-Text Decomposition and Perceptual CIELAB Bijection

Given an input RGB image $I_{\text{src}} \in \{0, \dots, 255\}^{H \times W \times 3}$ and a target style reference palette $Y_{\text{ref}} \in \{0, \dots, 255\}^{H_s \times W_s \times 3}$, Stage 0 invokes a multimodal VLM (`gemini-2.5-flash`, `gemini-2.5-pro`, or `gemini-3-flash-preview`) with deterministic decoding (temperature = 0.0, max_output_tokens = 256, and thinking_budget = 0 on Flash models) using the structured JSON prompt schema:

```
"You are the Stage-0 Visual-to-Text
Decomposer (GhP) for Agent-GhibliLock.
Scene metadata: {scene_summary}. Return
compact JSON with keys: {ghp_prompt:
<concise Studio Ghibli gouache style
prompt preserving emblem & diagonal
rail>, structural_retention_prior:
<float in [0.05, 0.15]>}"
```

The VLM returns both the natural-language style prompt (`ghp_prompt`) and a scalar structural retention prior $\hat{\eta}_{\text{VLM}} \in [0.05, 0.15]$ (`structural_retention_prior`), which directly parameterizes the downstream candidate synthesis harness ($\eta_{\text{struct}} = \hat{\eta}_{\text{VLM}}$ in live Vertex AI tracks, or $\eta_{\text{struct}} \in [0.04, 0.20]$ in controlled curriculum tracks) to produce a macro-registered candidate stylized asset $I_{\text{cand}} \in \{0, \dots, 255\}^{H \times W \times 3}$.

To decouple perceptual lightness and structural edges from chromaticity, all three images $\{I_{\text{src}}, I_{\text{cand}}, Y_{\text{ref}}\}$ are mapped into the CIE 1976 (L^*, a^*, b^*) color space under the CIE Standard Illuminant D65 white point $(X_n, Y_n, Z_n) = (0.95047, 1.00000, 1.08883)$. For normalized sRGB channels $c \in [0, 1]$, inverse companding (IEC 61966-2-1) yields linear light:

$$c_{\text{lin}} = \begin{cases} c/12.92, & \text{if } c \le 0.04045, \\ ((c + 0.055)/1.055)^{2.4}, & \text{otherwise.} \end{cases} \tag{1}$$

Linear RGB is projected to CIE XYZ via $M_{\text{RGB}\to\text{XYZ}}$ and transformed via the piecewise cubic root transfer function $f(t)$ with threshold $\delta = 6/29$:

$$f(t) = \begin{cases} t^{1/3}, & t > \delta^3, \\ \frac{t}{3\delta^2} + \frac{4}{29}, & \text{else,} \end{cases} \quad \begin{aligned} L^* &= 116 f(Y/Y_n) - 16, \\ a^* &= 500\,(f(X/X_n) - f(Y/Y_n)), \\ b^* &= 200\,(f(Y/Y_n) - f(Z/Z_n)). \end{aligned} \tag{2}$$

The inverse map LAB $\to$ sRGB applies the analytical inverse f^{-1}, multiplies by $M_{\text{XYZ}\to\text{RGB}} = M^{-1}_{\text{RGB}\to\text{XYZ}}$, clips out-of-gamut linear values to $[0, 1]$, applies forward gamma companding, and rounds to uint8, achieving round-trip error ≤ 1 LSB across $\{0, \dots, 255\}^3$.

B. Stage 1: Orthonormal Multi-Level 2D Haar Wavelet Decomposition and Structural Hashing

Let $L^* \in [0, 100]^{H \times W}$ denote a 2D luminance plane. When H or W is odd at any decomposition level, a single row or column is padded via edge replication prior to 2×2 analysis and cropped back to the exact target dimensions upon inverse synthesis.[1] For each 2×2 block $\mathbf{B}_{u,v} = [c_{00}\ c_{01};\ c_{10}\ c_{11}]$, the single-level orthonormal 2D Discrete Haar Wavelet Transform (DWT_2) [13], [14] computes:

$$\begin{bmatrix} LL(u,v) \\ LH(u,v) \\ HL(u,v) \\ HH(u,v) \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} c_{00} \\ c_{01} \\ c_{10} \\ c_{11} \end{bmatrix} = H_2 \mathbf{c}_{u,v}. \tag{3}$$

Because the 4×4 Haar matrix H_2 is symmetric and orthonormal ($H_2^T = H_2^{-1} = H_2$), the single-level inverse transform IDWT_2 is exact ($\mathbf{c}_{u,v} = H_2^T [LL, LH, HL, HH]^T$) and satisfies Parseval's identity $\|L^*\|_F^2 = \|LL_K\|_F^2 + \sum_{k=1}^{K} (\|LH_k\|_F^2 + \|HL_k\|_F^2 + \|HH_k\|_F^2)$ for any K-level recursive cascade on dyadic dimensions.

From the Level-1 detail subbands (LH_1, HL_1, HH_1) of L^*_{src}, we define the half-resolution structural edge energy map $E_{1/2}(u, v)$ and its $2\times$ nearest-neighbor upsampled, scale-normalized full-resolution map $E_{\text{full}}(L^*) \in [0, 1]^{H \times W}$:

$$\begin{aligned} E_{1/2}(u,v) &= \sqrt{LH_1(u,v)^2 + HL_1(u,v)^2 + \tfrac{1}{2} HH_1(u,v)^2}, \\ E_{\text{full}}(x,y) &= \frac{E_{1/2}(\lfloor x/2 \rfloor, \lfloor y/2 \rfloor)}{\max\left(\max_{u,v} E_{1/2}(u,v),\ 10^{-8}\right)}. \end{aligned} \tag{4}$$

Thresholding $E_{\text{full}}(L^*_{\text{src}})$ at $q_{0.75} = \max(\text{Quantile}_{0.75}(E_{\text{full}}),\ 0.08)$ yields the binary structural

[1] Exact Parseval energy conservation and zero-mean detail synthesis hold without boundary terms whenever $2^K \mid H$ and $2^K \mid W$ (e.g., 64×64, 128×128, 256×256, 512×512); for non-dyadic odd dimensions, edge replication introduces an $O(1/\min(H, W))$ boundary approximation.

edge/logo mask $M_{\text{HF}} \in \{0,1\}^{H\times W}$. To bind the source geometry into the C2PA manifest, M_{HF} is serialized as a contiguous row-major (C-order) uint8 byte buffer (`0x00`/`0x01`) concatenated with an 8-byte big-endian dimension trailer $\text{pack}(\texttt{">II"}, H, W)$ and hashed via SHA-256:

$$h_{\text{struct}} = \text{SHA256}\left(\text{bytes}_{\text{C-uint8}}(M_{\text{HF}}) \parallel \text{pack}(\texttt{">II"}, H, W)\right). \quad (5)$$

a) Structural Verification Certificate and Non-Circular Spatial Metrics: During closed-loop compilation, we monitor the global Structural Similarity Index ($\text{SSIM}_{\text{edge}}$) [5] between $E_1 = E_{\text{full}}(L^*_{\text{src}})$ and $E_2 = E_{\text{full}}(L^*_{\text{out}})$ with $c_1 = 0.01^2, c_2 = 0.03^2$:

$$\text{SSIM}_{\text{edge}}(L^*_{\text{src}}, L^*_{\text{out}}) = \frac{(2\mu_{E_1}\mu_{E_2} + c_1)(2\sigma_{E_1E_2} + c_2)}{(\mu^2_{E_1} + \mu^2_{E_2} + c_1)(\sigma^2_{E_1} + \sigma^2_{E_2} + c_2)}. \quad (6)$$

Because $\text{SSIM}_{\text{edge}}$ operates on the scale-normalized Level-1 Haar detail energy map E_{full}, it serves as an internal *compiler verification certificate* confirming that source detail subbands were retained across sRGB gamut clipping and Stage-3 DCT QIM watermarking. However, because E_{full} is invariant to global contrast scaling and local sign flips ($LH_1 \mapsto -LH_1$), relying solely on $\text{SSIM}_{\text{edge}}$ would be partially self-referential. We therefore pair $\text{SSIM}_{\text{edge}}$ in both the closed-loop stopping criterion and all benchmark evaluations with two non-circular spatial-domain metrics: (i) $\text{SSIM}_{\text{full}}$, computed directly over non-overlapping 4×4 spatial luminance patches ($L^*/100$, stride 4) without wavelets; and (ii) signed 3×3 Sobel gradient inner-product correlation $\rho_{\text{grad}} \in [-1, 1]$ evaluated over the interior structural edge locus M°_{HF}:

$$\rho_{\text{grad}}(L^*_{\text{src}}, L^*_{\text{out}}) = \frac{\sum_{(x,y)\in M^\circ_{\text{HF}}} \langle \nabla_{\text{Sobel}} L^*_{\text{src}}(x,y),\ \nabla_{\text{Sobel}} L^*_{\text{out}}(x,y)\rangle}{\|\nabla_{\text{Sobel}} L^*_{\text{src}}\|_{2,M^\circ_{\text{HF}}} \cdot \|\nabla_{\text{Sobel}} L^*_{\text{out}}\|_{2,M^\circ_{\text{HF}}} + 10^{-12}}, \quad (7)$$

which reverses sign ($\rho_{\text{grad}} \to -1$) under edge polarity inversions and penalizes 4×4 block boundary discontinuities.

C. Stage 2: 3D Sliced-Wasserstein Optimal Transport and Wavelet-Locked Subband Fusion

a) Step A: 3D CIELAB Sliced-Wasserstein Optimal Transport (SWOT): Let $X = \{x_i\}_{i=1}^N \subset \mathbb{R}^3$ ($N = HW$) denote the flat row-major sequence of CIELAB pixels of I_{cand}, and let $Y = \{y_j\}_{j=1}^N \subset \mathbb{R}^3$ denote the target style palette Y_{ref} (nearest-neighbor resampled to $H \times W$ when $H_s \times W_s \neq H \times W$). For any unit direction $\theta \in \mathbb{S}^2$, the 1D projected empirical Wasserstein-2 distance is $W_2(\theta_\# \mu_X, \theta_\# \mu_Y) = \left(\frac{1}{N}\sum_{r=1}^N (\langle x_{\pi_\theta(r)}, \theta\rangle - \langle y_{\sigma_\theta(r)}, \theta\rangle)^2\right)^{1/2}$, where $\pi_\theta, \sigma_\theta$ are stable sorting permutations with ties broken deterministically in lexicographical order of the flat 0-based row-major pixel index $i \in \{0, \ldots, N-1\}$ (`kind="mergesort"`).

To match the true quadratic Sliced-Wasserstein-2 metric [17], [18], we define both the in-sample training objective $\widehat{W}_2$ over the $D = 3M + 3 = 27$ orthonormal axes $\Theta_{\text{train}} = I_3 \cup \{Q_m\}_{m=1}^M$ ($M = 8$ Haar-uniform orthogonal frames $Q_m \in O(3)$ generated via positive-diagonal QR factorization $Z_m = Q_m R_m$, $Q_m \leftarrow Q_m \text{diag}(\text{sign}(\text{diag}(R_m)))$ from `seed=42`)[2] and the out-of-sample evaluation metric $\widehat{SW}_2^{\text{eval}}$ over $M_{\text{eval}} = 64$ independent uniform unit-sphere directions (`seed=999`) as root-mean-square (RMS) estimators:

$$\widehat{W}_2(\mu_X, \mu_Y) = \left(\frac{1}{D}\sum_{r=1}^D W_2^2(\theta_{r\#}\mu_X, \theta_{r\#}\mu_Y)\right)^{1/2},$$
$$\widehat{SW}_2^{\text{eval}}(\mu_X, \mu_Y) = \left(\frac{1}{M_{\text{eval}}}\sum_{s=1}^{M_{\text{eval}}} W_2^2(\tilde{\theta}_{s\#}\mu_X, \tilde{\theta}_{s\#}\mu_Y)\right)^{1/2}. \quad (8)$$

Step A initializes $X^{(0)}$ via coordinate-wise 1D quantile transport along $Q_0 = I_3$ with step size $\alpha_\tau \in [0.85, 0.97]$, iterates across $m = 1, \ldots, M$ orthonormal bases $Q_m \in O(3)$:

$$X^{(m)} = X^{(m-1)} + \alpha_\tau \sum_{d=1}^3 \left(\mathcal{Q}_{Y\cdot q_{m,d}}(\text{rank}(X^{(m-1)}q_{m,d})) - X^{(m-1)}q_{m,d}\right) q_{m,d}^T, \quad (9)$$

applies a half-step polish ($0.5\alpha_\tau$) over $\{I_3, Q_{M-1}, Q_M\}$, and enforces an explicit post-Step-A luminance mean re-anchoring shift:

$$\widetilde{L}^*_{\text{cand}}(x,y) \leftarrow \widetilde{L}^*_{\text{cand}}(x,y) + \left(\mu(L^*_{\text{ref}}) - \mu(\widetilde{L}^*_{\text{cand}})\right), \quad (10)$$

guaranteeing that $\mu(\widetilde{L}^*_{\text{cand}}) = \mu(L^*_{\text{ref}})$ holds identically entering Step B.

b) Step B: Global vs. Spatially Masked ROI Wavelet Detail Locking: Applying pointwise rank-order sorting in Step A aligns the 3D color distribution $\widetilde{I}_{\text{cand}} = (\widetilde{L}^*_{\text{cand}}, \widetilde{a}^*_{\text{cand}}, \widetilde{b}^*_{\text{cand}})$ but does not restore high-frequency source geometry and can introduce subtle 4×4 tile-boundary quantile steps in $LL_K(\widetilde{L}^*_{\text{cand}})$. Furthermore, in natural photographs with noisy backgrounds or skies, locking 100% of high-frequency wavelet details globally transfers photographic sensor grain into background regions that should exhibit smooth Studio Ghibli gouache brushwork.

Agent-GhibliLock therefore supports both **Global Wavelet Locking** ($M_{\text{ROI}}(x,y) \equiv 1$) and **Spatially Masked ROI Wavelet Locking**, where a smooth spatial weighting mask $M_{\text{ROI}}(x,y) \in [0,1]^{H\times W}$ is constructed from $E_{\text{full}}(L^*_{\text{src}})$ weighted by 5×5 Sobel directional coherence:

$$\mathcal{C}_{5\times 5}(x,y) = \frac{\sqrt{(\mathcal{B}_5 * G_x)^2 + (\mathcal{B}_5 * G_y)^2}}{\mathcal{B}_5 * \sqrt{G_x^2 + G_y^2} + 10^{-6}},$$
$$W_{\text{edge}}(x,y) = E_{\text{full}}(L^*_{\text{src}})(x,y) \cdot \mathcal{C}_{5\times 5}(x,y)^3. \quad (11)$$

Coherent structural contours, logos, and rails satisfy $\mathcal{C}_{5\times 5} \approx 1$ and are dilated by radius 2 px and feathered via a 3×3 box filter $\mathcal{B}_3$ to form $M_{\text{ROI}}(x,y) \in [0,1]$, whereas isolated background sensor grain has $\mathcal{C}_{5\times 5} \ll 1$ and receives weight 0. At each wavelet level $k \in \{1, \ldots, K\}$, M_{ROI} is downsampled to subband dimensions $(H/2^k) \times (W/2^k)$ via 2×2 block max-pooling to form $M_k(u,v) \in [0,1]$, ensuring every 2×2 Haar block overlapping the ROI receives full locking weight $\Gamma_{k,\tau}(u,v) = \gamma_\tau M_k(u,v)$. Step B blends source detail

[2] We sample from the full orthogonal group $O(3)$ rather than $SO(3)$ because 1D Wasserstein-2 distance is reflection-invariant: $W_2((-\theta)_\#\mu_X, (-\theta)_\#\mu_Y) = W_2(\theta_\#\mu_X, \theta_\#\mu_Y)$.

subbands $D_k(L^*_{\text{src}})$ with the smooth pre-sort candidate detail subbands $D_k(L^*_{\text{cand}})$ (eliminating Step A rank-sort staircasing):

$$\begin{aligned}\widehat{LL}_K &= \mathcal{S}_{3\times3}\Big((1-\eta_\tau)LL_K(\widetilde{L}^*_{\text{cand}}) + \eta_\tau \mathcal{Q}_{LL_K(L^*_{\text{ref}})}\Big(LL_K(\widetilde{L}^*_{\text{cand}})\Big)\Big),\\ \widehat{D}_k &= \Gamma_{k,\tau}\odot D_k(L^*_{\text{src}}) + (1-\Gamma_{k,\tau})\odot D_k(L^*_{\text{cand}}),\end{aligned} \tag{12}$$

for $D \in \{LH, HL, HH\}$ and $k \in \{1,\dots,K\}$, where $\mathcal{Q}_{LL_K(L^*_{\text{ref}})}$ is the exact tie-broken rank-order quantile bijection onto the $(H/2^K)\times(W/2^K)$ subband $LL_K(L^*_{\text{ref}})$ (which is a bijection of the empirical multiset preserving $\mu(\mathcal{Q}_{LL_K(L^*_{\text{ref}})}(Z)) = \mu(LL_K(L^*_{\text{ref}})) = 2^K\mu(L^*_{\text{ref}}))$ and $\mathcal{S}_{3\times3}(Z) = 0.65Z + 0.35(\mathcal{B}_3 * Z) + (\mu(Z) - \mu(0.65Z + 0.35(\mathcal{B}_3 * Z)))$ is a mean-preserving 3×3 coarse-subband smoothing operator.

c) Decoupled $(\tau_\gamma, \tau_\alpha)$ Closed-Loop Controller: Synthesizing $\widehat{L}^*_{\text{raw}} = \text{IDWT}_2(\widehat{LL}_K, \{\widehat{D}_k\}_{k=1}^K)$ can inflate global luminance variance when L^*_{src} has extreme contrast, which would increase $\widehat{W}_2^{\text{StepB}}$. Crucially, if a single coupled index τ simultaneously increased γ_τ whenever $\widehat{W}_2^{\text{StepB}} > \tau_{\text{color}}$, injecting *more* high-frequency source detail energy would move $\widehat{W}_2^{\text{StepB}}$ in the wrong direction. Agent-GhibliLock therefore maintains two decoupled schedule indices $\tau_\gamma, \tau_\alpha \in \{0,1,2,3\}$ over $\gamma \in \{0.78, 0.88, 0.94, 0.98\}$, $\alpha \in \{0.85, 0.90, 0.94, 0.97\}$, $\eta \in \{0.00, 0.45, 0.75, 0.95\}$, and variance-matching weight $\nu \in \{0.00, 0.72, 0.90, 0.98\}$:

1) Prior to clamping, if $\tau_\alpha > 0$ and $\sigma(\widehat{L}^*_{\text{raw}}) > \sigma(\widetilde{L}^*_{\text{cand}})$, apply the affine variance-regularization map $\widehat{L}^*_{\text{raw}} \leftarrow \mu(\widetilde{L}^*_{\text{cand}}) + \kappa_{\tau_\alpha}(\widehat{L}^*_{\text{raw}} - \mu(\widehat{L}^*_{\text{raw}}))$ with $\kappa_{\tau_\alpha} = (1-\nu_{\tau_\alpha}) + \nu_{\tau_\alpha}\sigma(\widetilde{L}^*_{\text{cand}})/\sigma(\widehat{L}^*_{\text{raw}})$. Because positive affine scaling around the mean cancels identically in both E_{full} (Eq. 4) and ρ_{grad} (Eq. 7), this step contracts $\widehat{W}_2^{\text{StepB}}$ while leaving $\text{SSIM}_{\text{edge}}$ and ρ_{grad} 100% invariant.
2) Clamp $\widehat{L}^*_{\text{clip}} = \text{clip}(\widehat{L}^*_{\text{raw}}, 0, 100)$ and apply interior DC re-anchoring over unsaturated pixels $\mathcal{I}^\circ = \{(x,y) : \widehat{L}^*_{\text{clip}}(x,y) \in (1,99)\}$ via $\widehat{L}^*(x,y) = \text{clip}(\widehat{L}^*_{\text{clip}}(x,y) + \frac{HW}{|\mathcal{I}^\circ|}\Delta_\mu \mathbf{1}_{(x,y)\in\mathcal{I}^\circ}, 0, 100)$ so $\mu(\widehat{L}^*) = \mu(\widetilde{L}^*_{\text{cand}}) = \mu(L^*_{\text{ref}})$.
3) If $\text{SSIM}_{\text{edge}} < \tau_{\text{struct}}$ or $\rho_{\text{grad}} < 0.88$, increment $\tau_\gamma \leftarrow \min(3, \tau_\gamma + 1)$ and $\tau_\alpha \leftarrow \min(3, \tau_\alpha + 1)$; else if $\widehat{W}_2^{\text{StepB}} > \tau_{\text{color}}$, hold τ_γ fixed and increment only $\tau_\alpha \leftarrow \min(3, \tau_\alpha + 1)$.

Theorem 1 (Exact Spatial Mean Luminance Conservation under Spatially Masked Haar Detail Locking)**.** *Let $L^*_{src}, \widetilde{L}^*_{cand}, L^*_{ref} \in \mathbb{R}^{H\times W}$ have dyadic dimensions ($2^K \mid H, W$), where $\widetilde{L}^*_{cand}$ has been mean-anchored via Eq. (10) such that $\mu(\widetilde{L}^*_{cand}) = \mu(L^*_{ref})$. For any K-level Haar decomposition and any spatially varying detail mask $M_k(u,v) \in [0,1]^{(H/2^k)\times(W/2^k)}$, the pre-clamping wavelet-locked luminance $\widehat{L}^*_{raw} = \text{IDWT}_2(\widehat{LL}_K, \{\widehat{LH}_k, \widehat{HL}_k, \widehat{HH}_k\}_{k=1}^K)$ satisfies:*

$$\mu(\widehat{L}^*_{raw}) = \frac{1}{HW}\sum_{x=0}^{H-1}\sum_{y=0}^{W-1}\widehat{L}^*_{raw}(x,y) = 2^{-K}\mu(\widehat{LL}_K) = \mu(\widetilde{L}^*_{cand}) = \mu(L^*_{ref}). \tag{13}$$

Moreover, let $\widehat{L}^*_{clip} = \text{clip}(\widehat{L}^*_{raw}, 0, 100)$ and let $\mathcal{I}^\circ = \{(x,y) : \widehat{L}^*_{clip}(x,y) \in (1,99)\}$ denote the interior unsaturated pixel locus. If $\mathcal{I}^\circ \neq \emptyset$ and the clamping drift $\Delta_\mu = \mu(L^*_{ref}) - \mu(\widehat{L}^*_{clip})$ satisfies $|\Delta_\mu| \cdot (HW/|\mathcal{I}^\circ|) \le 1$, the piecewise interior DC restoration map $\widehat{L}^*(x,y) = \text{clip}\Big(\widehat{L}^*_{clip}(x,y) + \frac{HW}{|\mathcal{I}^\circ|}\Delta_\mu \mathbf{1}_{(x,y)\in\mathcal{I}^\circ}, 0, 100\Big)$ restores $\mu(\widehat{L}^*) = \mu(L^*_{ref})$ identically without boundary resaturation.

Proof. At any single inverse Haar synthesis level k, each 2×2 block $\mathbf{c}_{u,v} \in \mathbb{R}^4$ is reconstructed via $\mathbf{c}_{u,v} = H_2^T[LL_{k-1}(u,v), \widehat{LH}_k(u,v), \widehat{HL}_k(u,v), \widehat{HH}_k(u,v)]^T$. Multiplying on the left by $\mathbf{1}_4^T = [1,1,1,1]$ yields:

$$\mathbf{1}_4^T\mathbf{c}_{u,v} = (\mathbf{1}_4^T H_2^T)\begin{bmatrix}LL_{k-1}(u,v)\\ \widehat{LH}_k(u,v)\\ \widehat{HL}_k(u,v)\\ \widehat{HH}_k(u,v)\end{bmatrix} = \begin{bmatrix}2 & 0 & 0 & 0\end{bmatrix}\begin{bmatrix}LL_{k-1}(u,v)\\ \widehat{LH}_k(u,v)\\ \widehat{HL}_k(u,v)\\ \widehat{HH}_k(u,v)\end{bmatrix} = 2\,LL_{k-1}(u,v), \tag{14}$$

regardless of the values of $(\widehat{LH}_k(u,v), \widehat{HL}_k(u,v), \widehat{HH}_k(u,v)) = \Gamma_{k,\tau}(u,v)D_k(L^*_{\text{src}})(u,v) + (1-\Gamma_{k,\tau}(u,v))D_k(L^*_{\text{cand}})(u,v)$. Summing over all (u,v) and inducting from $k = K$ down to $k = 0$ gives $\sum_{x,y}\widehat{L}^*_{\text{raw}}(x,y) = 2^K\sum_{u,v}\widehat{LL}_K(u,v)$. Because the rank-order bijection $\mathcal{Q}_{LL_K(L^*_{\text{ref}})}$ is a permutation of $LL_K(L^*_{\text{ref}})$ on equal $(H/2^K)\times(W/2^K)$ grids and $\mathcal{S}_{3\times3}$ explicitly re-centers its output mean, $\mu(\widehat{LL}_K) = (1-\eta_\tau)2^K\mu(\widetilde{L}^*_{\text{cand}}) + \eta_\tau 2^K\mu(L^*_{\text{ref}}) = 2^K\mu(L^*_{\text{ref}})$. Dividing by HW yields Eq. (13). Finally, when $|\Delta_\mu|(HW/|\mathcal{I}^\circ|) \le 1$, every $(x,y) \in \mathcal{I}^\circ$ satisfies $\widehat{L}^*_{\text{clip}}(x,y) + \frac{HW}{|\mathcal{I}^\circ|}\Delta_\mu \in (0,100)$, so the outer clip is inactive and summing over $\{0,\dots,H-1\}\times\{0,\dots,W-1\}$ yields $\mu(\widehat{L}^*) = \mu(\widehat{L}^*_{\text{clip}}) + \Delta_\mu = \mu(L^*_{\text{ref}})$. □

Theorem 2 (Exact Deterministic RMS Sliced-Wasserstein Perturbation Bound)**.** *Let $\widetilde{I}_{cand} = (\widetilde{L}^*_{cand}, \widetilde{a}^*_{cand}, \widetilde{b}^*_{cand})$ and $\widehat{I} = (\widehat{L}^*, \widetilde{a}^*_{cand}, \widetilde{b}^*_{cand})$ share identical transported chromaticity channels $(\widetilde{a}^*_{cand}, \widetilde{b}^*_{cand})$. Over any collection of $M+1$ complete orthonormal frames $\{Q_0, Q_1, \dots, Q_M\} \subset O(3)$ ($D = 3(M+1)$ unit projection axes), the root-mean-square Sliced-Wasserstein-2 distance $\widehat{W}_2$ (Eq. 8) satisfies the exact deterministic bound:*

$$\left|\widehat{W}_2(\mu_{\widehat{I}}, \mu_{Y_{\text{ref}}}) - \widehat{W}_2(\mu_{\widetilde{I}_{\text{cand}}}, \mu_{Y_{\text{ref}}})\right| \le \widehat{W}_2(\mu_{\widehat{I}}, \mu_{\widetilde{I}_{\text{cand}}}) \le \frac{1}{\sqrt{3HW}}\left\|\widehat{L}^* - \widetilde{L}^*_{cand}\right\|_F. \tag{15}$$

Proof. Because $\widehat{W}_2$ is an L_2 root-mean-square norm of 1D Wasserstein-2 distances across $r \in \{1,\dots,D\}$, the reverse triangle inequality implies $|\widehat{W}_2(\mu_{\widehat{I}}, \mu_{Y_{\text{ref}}}) - \widehat{W}_2(\mu_{\widetilde{I}_{\text{cand}}}, \mu_{Y_{\text{ref}}})| \le \widehat{W}_2(\mu_{\widehat{I}}, \mu_{\widetilde{I}_{\text{cand}}})$. For each axis $\theta_r = (\theta_{r,L^*}, \theta_{r,a^*}, \theta_{r,b^*}) \in \mathbb{S}^2$, the Hardy-Littlewood-Pólya rearrangement inequality implies that sorting is a non-expansive operator in $\ell_2(\mathbb{R}^N)$:

$$W_2^2(\theta_{r\#}\mu_{\widehat{I}}, \theta_{r\#}\mu_{\widetilde{I}_{\text{cand}}}) \le \frac{1}{N}\sum_{i=1}^N\langle\widehat{I}_i - \widetilde{I}_{\text{cand},i}, \theta_r\rangle^2 = \frac{(\theta_{r,L^*})^2}{HW}\left\|\widehat{L}^* - \widetilde{L}^*_{\text{cand}}\right\|_F^2, \tag{16}$$

since $\widehat{a}^* - \widetilde{a}^*_{\text{cand}} = 0$ and $\widehat{b}^* - \widetilde{b}^*_{\text{cand}} = 0$. Averaging over the $D = 3(M+1)$ axes partitioned into $M+1$ orthogonal matrices $Q_m \in O(3)$, the three columns of each Q_m form an orthonormal basis of $\mathbb{R}^3$, so the first row of every Q_m has unit Euclidean norm: $\sum_{j=1}^3(Q_{m,1j})^2 = 1$. Consequently,

$\frac{1}{D}\sum_{r=1}^{D}(\theta_{r,L^*})^2 = \frac{1}{3(M+1)}\sum_{m=0}^{M} 1 = \frac{1}{3}$ holds *deterministically and identically* for all $M \geq 0$. Taking square roots yields Eq. (15). □

D. Temporal Video GOP Extension: Motion-Compensated and Motion-Gated Wavelet EMA

For an 8-frame Group of Pictures (GOP) $\{I_{\text{src}}^{(t)}, I_{\text{cand}}^{(t)}\}_{t=1}^{T}$, independent diffusion synthesis causes severe frame-to-frame luminance flicker $\sigma^2_{\text{flicker}} = \text{Var}_t(\overline{L}^{*(t)} - \overline{L}^{*(t-1)})$, whereas unwarped temporal EMA filtering ghosts across moving objects. Agent-GhibliLock estimates the integer-tile translation $(\Delta u_t, \Delta v_t)$ between $LL_K(L_{\text{src}}^{*(t-1)})$ and $LL_K(L_{\text{src}}^{*(t)})$ by minimizing zero-mean cross-correlation MSE (accepting non-zero shifts only when MSE drops by $\geq 35\%$), computes the scale-normalized motion innovation $\Delta_{\text{mot}}^{(t)}(u,v) = 2^{-K}|LL_K(L_{\text{src}}^{*(t)})(u,v) - \mathcal{T}_{\Delta u_t, \Delta v_t} LL_K(L_{\text{src}}^{*(t-1)})(u,v)|$, and applies the translation-compensated, spatially gated coarse-subband update:

$$\beta_t(u,v) = \begin{cases} \beta \exp\left(-\dfrac{(\Delta_{\text{mot}}^{(t)}(u,v))^2}{2\sigma_{\text{mot}}^2}\right), & \Delta_{\text{mot}}^{(t)}(u,v) > \sigma_{\text{mot}}, \\ \beta, & \text{otherwise}, \end{cases}$$

$$\overline{LL}_K^{(t)} = \beta_t \odot \mathcal{T}_{\Delta u_t, \Delta v_t} \overline{LL}_K^{(t-1)} + (1-\beta_t) \odot LL_K(\widetilde{L}_{\text{cand}}^{*(t)}), \tag{17}$$

with $\beta = 0.78, \sigma_{\text{mot}} = 35.0$, followed by mean re-anchoring $\overline{LL}_K^{(t)} \leftarrow \overline{LL}_K^{(t)} + (2^K \mu(L_{\text{ref}}^*) - \mu(\overline{LL}_K^{(t)}))$. Per-frame SHA-256 digests $\{h_1, \ldots, h_T\}$ are folded pairwise into a binary Merkle root R_{GOP} stored in the container manifest.

E. Stage 3: Dual-Layer Provenance—Mid-Band 2D-DCT QIM Watermark and C2PA-Inspired JUMBF (`caBX`)

a) Mid-Band 2D-DCT Quantization-Index-Modulation (QIM) Watermark: To survive metadata chunk stripping, Stage 3 embeds a $N_w = 64$-bit cryptographic payload $\mathbf{w} \in \{0,1\}^{64}$ (derived from $\text{SHA256}(h_{\text{struct}} \,\|\, \text{model_ids})[:8]$) into the floating-point ITU-R BT.601 luminance channel $Y = 0.299R + 0.587G + 0.114B \in [0, 255]$ partitioned into $B = \lfloor H/8 \rfloor \lfloor W/8 \rfloor$ non-overlapping 8×8 blocks. Each shifted block $\mathbf{B}_b - 128$ is transformed via the orthonormal 8×8 Type-II DCT matrix C_8: $\mathbf{F}_b = C_8(\mathbf{B}_b - 128)C_8^T$. Crucially, we never modify the DC $(0,0)$ coefficient, restricting embedding strictly to mid-band AC coordinates under two supported layouts:

- **Standard 4-Mode AC Layout ($M_{\text{coord}} = 4$):** $\mathcal{M}_4 = \{(2,3),(3,2),(1,3),(3,1)\}$, yielding $R = \lfloor 4B/64 \rfloor = \lfloor B/16 \rfloor$ votes per bit ($R = 4$ at 64×64, $R = 16$ at 128×128, $R = 64$ at 256×256, $R = 256$ at 512×512).
- **Odd-Repetition 5-Mode AC Layout ($M_{\text{coord}} = 5$):** $\mathcal{M}_5 = \mathcal{M}_4 \cup \{(2,2)\}$, yielding $R = 5B/64 = 5$ strictly odd majority votes per bit at 64×64 with $\gcd(5, 64) = 1$, eliminating even-vote ties and dispersing bit assignments across both rows and columns.

For block $b \in \{0, \ldots, B-1\}$ and mode $m \in \{0, \ldots, M_{\text{coord}} - 1\}$, bit index $i = \phi(b,m) = (M_{\text{coord}} b + m) \bmod 64$ (or row-shifted $\phi_{\text{row}}(b,m) = (M_{\text{coord}} b + 4r_b + m) \bmod 64$ where $r_b = \lfloor b / \lfloor W/8 \rfloor \rfloor$ to break vertical column alignment when $16 \mid \lfloor W/8 \rfloor$) is embedded with step size $\Delta_{\text{QIM}} = 14.0$ and optional keyed pseudo-random dither $d_{b,m}(K_{\text{secret}}) \in [0, \Delta_{\text{QIM}})$ [21]:

$$\widetilde{F}_b(u_m, v_m) = \left\lfloor \frac{F_b(u_m, v_m) - \delta_{b,m}(w_i)}{\Delta_{\text{QIM}}} + \frac{1}{2} \right\rfloor \Delta_{\text{QIM}} + \delta_{b,m}(w_i),$$
$$\delta_{b,m}(w_i) = d_{b,m}(K_{\text{secret}}) + \frac{\Delta_{\text{QIM}}}{2} w_i. \tag{18}$$

The spatial luminance increment $\Delta Y = \text{IDCT}_8(\widetilde{\mathbf{F}}) - Y$ is added equally across (R,G,B) and rounded/clamped to $\{0, \ldots, 255\}$. During extraction, each block/mode contributes either a binary hard vote $\mathbf{1}\{\text{dist}(F_b^{\text{rx}}, \Lambda_1) < \text{dist}(F_b^{\text{rx}}, \Lambda_0)\} \in \{0,1\}$ or a **clipped soft coset vote** $v_{b,m} = \text{clip}\left(1 - \frac{\text{dist}(F_b^{\text{rx}}(u_m,v_m), \Lambda_1)}{\Delta_{\text{QIM}}/2}, 0, 1\right) \in [0,1]$, ensuring no single tampered block can contribute a vote outside $[0,1]$.

Theorem 3 (QIM Coset Minimum-Distance Margin with Exact Mid-Band AC Rounding Bounds and Blind Tamper Localization)**.** *Let $\widetilde{F}_b(u_m, v_m) \in \Lambda_{w_i}$ be a quantized mid-band AC DCT coefficient ($(u_m, v_m) \in \mathcal{M}_4$ or $\mathcal{M}_5$) before* uint8 *export. Upon adding ΔY equally to (R,G,B), rounding to $\mathbb{Z}$, clamping to $[0,255]$, and applying an external channel perturbation, the received DCT coefficient is $F_b^{rx}(u_m, v_m) = \widetilde{F}_b(u_m, v_m) + \eta_b^{export}(u_m, v_m) + \eta_b^{chan}(u_m, v_m)$.*

1) ***Exact Mid-Band AC ℓ_1 Rounding Bounds and Cross-Channel Export Self-Noise:*** *Since $\text{dist}(\Lambda_0, \Lambda_1) = \Delta_{QIM}/2$, a single coefficient decodes error-free whenever $|\eta_b^{export} + \eta_b^{chan}| < \Delta_{QIM}/4 = 3.5$. For interior pixels ($c + \Delta Y \in [0,255]$), adding ΔY equally to integer $R,G,B \in \mathbb{Z}$ produces identical fractional parts across all three channels, so $\epsilon_R = \epsilon_G = \epsilon_B = \epsilon_Y \in [-0.5, 0.5]$ with $|\epsilon_Y| \leq 0.5$ LSB. While the unwatermarked DC $(0,0)$ mode has ℓ_1 norm $\|c_0\|_1 = \sqrt{8}$ (worst-case bound 4.0), the 1D Type-II DCT AC basis vectors c_1, c_2, c_3 satisfy $\|c_1\|_1 = \|c_3\|_1 = \frac{1}{2}\csc(\pi/16) \approx 2.5629$ and $\|c_2\|_1 = \sqrt{4 + 2\sqrt{2}} \approx 2.6131$, yielding deterministic worst-case interior rounding bounds strictly below the QIM Voronoi radius $\Delta_{QIM}/4 = 3.5$:*

$$\max_{(u_m,v_m)\in\mathcal{M}_4} |\eta_b^{export}(u_m, v_m)| \leq \tfrac{1}{2}\|c_2\|_1\|c_3\|_1 \approx 3.3486 < 3.5,$$
$$\max_{(u_m,v_m)\in\mathcal{M}_5} |\eta_b^{export}(u_m, v_m)| \leq \tfrac{1}{2}\|c_2\|_1^2 = 2 + \sqrt{2} \approx 3.4142 < 3.5. \tag{19}$$

Hence interior uint8 *rounding alone can never flip a single watermarked AC coefficient ($p_e^{export} = 0$). Under uniform quantization noise $\epsilon_Y \sim \text{Unif}[-0.5, 0.5]$, orthonormality of C_8 yields theoretical interior rounding variance $\sigma^2_{round} = 1/12 \approx 0.0833$ (RMS ≈ 0.2887; empirical $\sigma^2_{round+clamp} = 0.0821 \ll 12.25$).*

2) ***R-Fold Repetition Voting:*** *Under zero-mean sub-Gaussian total perturbation with effective variance proxy $\sigma^2_{eff} = \sigma^2_{round+clamp} + \sigma^2_\eta$ across spatially disjoint 8×8 blocks, the single-coefficient flip probability satisfies $p_e \leq 2\exp(-\Delta^2_{QIM}/(32\sigma^2_{eff}))$. For odd R, majority voting yields $P_{bit} = \sum_{k=(R+1)/2}^{R} \binom{R}{k} p_e^k (1-p_e)^{R-k}$ (satisfying $P_{bit} \leq 10p_e^3$ for all $p_e \in [0, 0.5]$ at $R = 5$), whereas even R includes the tie-breaking term $\frac{1}{2}\binom{R}{R/2} p_e^{R/2}(1-p_e)^{R/2}$.*

3) ***Combinatorial Guarantee for Blind Coset Tamper Localization:*** *Let $\mathcal{B}_i = \{b : \exists m, \phi(b,m) = i\}$ denote the R distinct 8×8 blocks carrying bit i, and let $\mathcal{T} \subset \{0, \ldots, B-1\}$ denote the set of spatially tampered blocks. Whenever untampered blocks $b \notin \mathcal{T}$ satisfy $|\eta_b^{export} + \eta_b^{chan}| < \Delta_{QIM}/4$ and $|\mathcal{T} \cap \mathcal{B}_i| \le \lfloor (R-1)/2 \rfloor$ for every bit $i \in \{0, \ldots, 63\}$, both hard majority voting and clipped-soft voting ($v_{b,m} \in [0,1]$) recover $\hat{\mathbf{w}} = \mathbf{w}$ with zero errors regardless of tamper amplitude, enabling blind block-level tamper localization via $\max_m \mathrm{dist}(F_b^{rx}(u_m, v_m), \Lambda_{\hat{w}_i}) > \Delta_{QIM}/4$ (or ≥ 3 modes exceeding $0.4\Delta_{QIM}$ in JPEG-resilient mode).*

Proof. Since $\eta_b^{\text{export}}(u,v) = \sum_{x=0}^{7}\sum_{y=0}^{7} c_u(x)c_v(y)\epsilon_Y(x,y)$, Hölder's inequality with $\|\epsilon_Y\|_\infty \le 0.5$ gives $|\eta_b^{\text{export}}(u,v)| \le \frac{1}{2}\|c_u\|_1\|c_v\|_1$. Direct evaluation of $\|c_k\|_1 = \sum_{x=0}^{7}\frac{1}{2}|\cos(\frac{(2x+1)k\pi}{16})|$ yields $\|c_1\|_1 = \|c_3\|_1 = \frac{1}{2}\csc(\pi/16) \approx 2.5629$ and $\|c_2\|_1 = \sqrt{4+2\sqrt{2}} \approx 2.6131$, establishing Eq. (19) and (1). Part (2) follows from Hoeffding's sub-Gaussian tail bound on the Voronoi half-width $\Delta_{\text{QIM}}/4$ and binomial counting over the R disjoint blocks in $\mathcal{B}_i$ (since $f(p_e) = 10p_e^3 - [10p_e^3(1-p_e)^2 + 5p_e^4(1-p_e) + p_e^5] = p_e^3(15p_e - 15p_e^2 + 4p_e^2) = p_e^4(15 - 11p_e) \ge 0$ for all $p_e \in [0,1]$). For (3), because each untampered block $b \in \mathcal{B}_i \backslash \mathcal{T}$ casts a valid vote for w_i and each tampered block $b \in \mathcal{T} \cap \mathcal{B}_i$ is clipped to $[0,1]$, at least $R - \lfloor (R-1)/2 \rfloor = \lceil (R+1)/2 \rceil > R/2$ blocks vote correctly for w_i, guaranteeing $\hat{w}_i = w_i$. □

b) Compact C2PA-Inspired ISO/IEC 19566-5 JUMBF (`caBX`) PNG Profile: To provide sub-millisecond container-level attestation without external ASN.1/X.509/COSE dependencies, Stage 3 constructs a compact *C2PA-inspired ISO/IEC 19566-5 JUMBF microservice profile* [7], [8] and injects it as an ancillary `caBX` chunk immediately after the 25-byte PNG `IHDR` chunk (13-byte `IHDR` payload plus 12-byte chunk framing: 4B length, 4B tag, 4B IEEE 802.3 CRC-32) [9]. The outer BMFF superbox (`jumb`) encapsulates: (1) a Description Box (`jumd`) carrying the 16-byte C2PA UUID (`63327061-0011-0010-8000-00aa00389b71`), requestable toggle `0x03`, and null-terminated label `c2pa.assertions.wavelet_style_lock`; (2) a canonical UTF-8 JSON Assertion Box (`json`) binding `source_structural_mask_sha256`, `output_pixel_sha256`, all four Wasserstein checkpoints ($\widehat{W}_2^{\text{StepA}}, \widehat{W}_2^{\text{StepB}}, \widehat{W}_2^{\text{out}}, \widehat{SW}_2^{\text{eval}}$), $(\gamma_\tau, \alpha_\tau)$, and `dct_watermark_hex`; and (3) a keyed HMAC-SHA256 Signature Box (`c2sh`, 32B digest over `jumd` ∥ `json`).[3]

[3]In full enterprise deployments requiring public-key third-party validation via `c2pa-rs`, the `json` box maps 1-to-1 to CBOR-encoded C2PA assertion/claim superboxes (each a BMFF `jumb` superbox whose inner `jumd` Description Box carries the C2PA content-type/label `c2as` or `c2cl`) nested inside a manifest store (`c2pa`) and active manifest (`c2ma`) `jumb` hierarchy signed via COSE_Sign1 X.509 certificates.

III. Experimental Evaluation

A. Experimental Setup, Baselines B1–B7, and Evaluation Metrics

We evaluate Agent-GhibliLock across three complementary testbeds:

1) **Stratified $N = 600$ Benchmark (`run_live_benchmark.py`):** Five workload strata at 64×64 ($B = 64$ DCT blocks): S_1 ($n = 150$, apparel diamond emblems), S_2 ($n = 150$, perspective rails and pillars), S_3 ($n = 100$, foliage and gouache skies), S_4 ($n = 100$, 8-frame video GOP sequences under $[-24, +28]$ LSB temporal exposure flicker), and S_5 ($n = 100$, four disjoint $n = 25$ adversarial provenance regimes A_1, A_2, A_{3a}, A_{3b}).
2) **Live Multi-Model Vertex AI Evaluation (`run_live_gemini_multimodel_eval.py`):** Live Stage-0 API calls to `gemini-2.5-flash`, `gemini-2.5-pro`, and `gemini-3-flash-preview` in `project="<gcp-project-id>"` (10 scenes $\times 128 \times 128$ per model), where each model's returned `structural_retention_prior` $\hat{\eta}_{\text{VLM}} \in [0.05, 0.15]$ directly drives candidate synthesis ($\eta_{\text{struct}} = \hat{\eta}_{\text{VLM}}$), alongside 5 controlled structural drift curriculum tiers (C1–C5, $\eta_{\text{struct}} \in [0.04, 0.20]$).
3) **Real Natural Photographic Benchmark (64^2–512^2):** Multi-resolution evaluation on real photographs `grace_hopper.jpg` ($64 \times 64, 128 \times 128, 256 \times 256, 512 \times 512$) and `Minduka_Present_Blue_Pack.png` ($128 \times 128$).

To isolate deterministic compiler convergence from stochastic diffusion seed variance across the $N = 600$ benchmark, candidate generation and all 9 comparative paradigms are parameterized explicitly in `run_live_benchmark.py`:

- **B1 (Unconstrained Prompt-Only):** Upstream text-driven synthesis with low structural retention ($\eta_{\text{struct}} \in [0.05, 0.08]$) and unaligned color drift (1419.9 ms mean latency including upstream API/synthesis time).
- **B2 ($3\times$ VLM-as-a-Judge Retry):** Three iterative VLM critique-and-regenerate rounds (4377.2 ms), which modestly reduce color drift ($\widehat{W}_2^{\text{out}} : 21.99 \to 19.44$) but cannot recover fine-scale phase across the prose bottleneck ($\text{SSIM}_{\text{edge}} = 0.0995$).
- **B3 ($M = 4$ and $M = 8$ Global 3D SWOT Only):** Applies Step-A 3D CIELAB Sliced-Wasserstein Optimal Transport (19.7 ms and 30.6 ms local CPU latency) without Step-B wavelet detail locking or Stage-3 provenance.
- **B4 (Post-Hoc C2PA Wrapper):** Wraps the unconstrained candidate in a `caBX` JUMBF manifest (3.2 ms) without wavelet locking or DCT watermarking.
- **B5 (WCT^2 Wavelet Style Transfer [12]):** Preserves Haar detail skips (LH_k, HL_k, HH_k) while applying 1D second-order Gaussian moment matching (μ, σ) on LL_K and chrominance (142.8 ms).

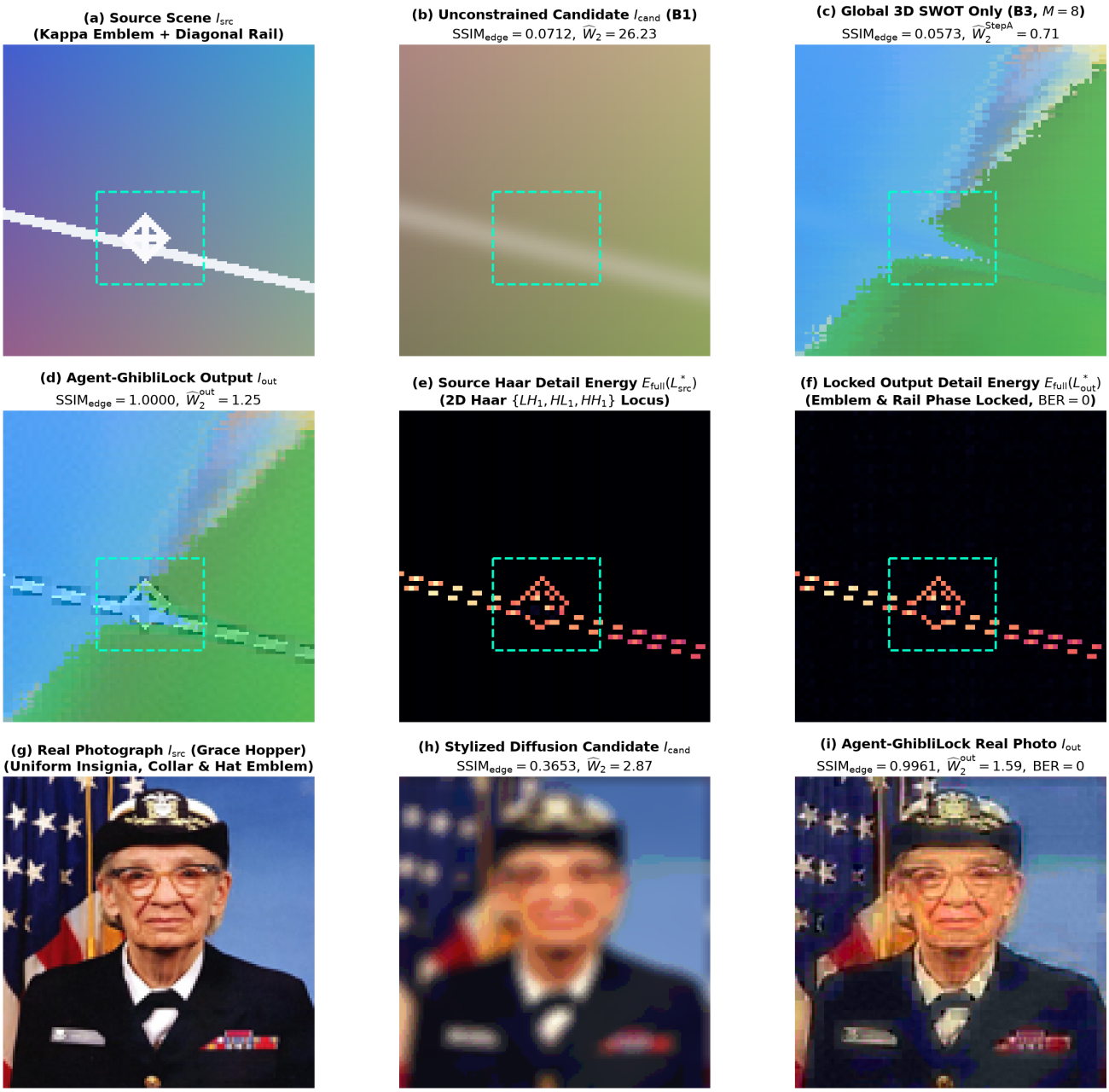


Fig. 1. Nine-panel qualitative and spectral raster comparison rendered by `run_live_benchmark.py` to both `plots/fig_qualitative_ghiblilock_comparison.png` and `fig_qualitative_ghiblilock_comparison.png`. **Rows 1–2 (Synthetic Emblem & Perspective Rail Scene,** 128×128**):** (a) Source scene I_{src}; (b) Unconstrained candidate I_{cand} (**B1**, $\mathrm{SSIM}_{edge} = 0.0712$, $\widehat{W}_2 = 26.23$); (c) Global 3D SWOT only (**B3**, $M = 8$, $\mathrm{SSIM}_{edge} = 0.0573$, $\widehat{W}_2^{StepA} = 0.71$); (d) **Agent-GhibliLock** output I_{out} ($\mathrm{SSIM}_{edge} = 1.0000$, $\widehat{W}_2^{out} = 1.25$); (e–f) Source vs. locked Haar Level-1 detail energy maps E_{full} (BER $= 0$). **Row 3 (Real Natural Photograph `grace_hopper.jpg`,** 128×128**):** (g) Source Naval portrait with hat insignia, eyeglasses, and collar stars; (h) Stylized diffusion candidate I_{cand} ($\mathrm{SSIM}_{edge} = 0.3653$, $\widehat{W}_2 = 2.87$); (i) **Agent-GhibliLock** output I_{out} ($\mathrm{SSIM}_{edge} = 0.9961$, $\widehat{W}_2^{out} = 1.59$, BER $= 0$).

- **B6 (Image-Conditioned SDEdit [3] at strength** $s = 0.55$**):** Initializes synthesis from a noisy source latent (1390.5 ms), partially retaining coarse color ($\widehat{W}_2^{out} = 14.17$) while smoothing 2 px emblem strokes ($\mathrm{SSIM}_{edge} = 0.0672$).
- **B7 (ControlNet** $8\times$**-VAE Adapter [4]):** Conditions generation on an $8\times$-downsampled/upsampled latent VAE spatial bottleneck paired with Step-A 3D SWOT ($M = 8$, 1652.1 ms total latency); Step-A SWOT aligns the palette ($\widehat{W}_2^{StepA} = 0.71, \widehat{W}_2^{out} = 3.20$), isolating how the $8\times$ latent VAE bottleneck preserves coarse gradient orientation ($\rho_{grad} = 0.8948$) yet blurs 2 px emblem strokes

($\text{SSIM}_{\text{edge}} = 0.2325$).

- **Agent-GhibliLock (Global and ROI-Masked):** Our full 4-stage compiler (116.2 ms and 130.5 ms mean local CPU latency).

B. Stratified $N = 600$ Benchmark and Multi-Stage Ablations

Table I and Table II summarize the $N = 600$ benchmark results:

- **Resolving the Structure–Color Pareto Trade-Off:** Pure 3D SWOT (**B3**, $M = 8$) achieves the lowest chromaticity distance ($\widehat{W}_2^{\text{out}} = 0.71, \widehat{SW}_2^{\text{eval}} = 0.82$) but collapses structural edge similarity ($\text{SSIM}_{\text{edge}} = 0.0186, \rho_{\text{grad}} = 0.4367$). Conversely, WCT^2 (**B5**) achieves 100.0% structural pass rate ($\text{SSIM}_{\text{edge}} = 0.9992, \rho_{\text{grad}} = 0.9825$) via 1D moment matching, but its joint structure-and-color pass rate is 0.0% because it fails to align the 3D gouache palette ($\widehat{W}_2^{\text{out}} = 9.68 > \tau_{\text{color}} = 4.5$). ControlNet (**B7**) preserves coarse gradient orientation ($\rho_{\text{grad}} = 0.8948$) and moderate palette alignment ($\widehat{W}_2^{\text{out}} = 3.20$), but its $8\times$ latent VAE bottleneck blurs 2 px emblem strokes ($\text{SSIM}_{\text{edge}} = 0.2325$, 0.0% pass rate). **Agent-GhibliLock (Global)** achieves 100.0% on ($\text{SSIM}_{\text{edge}} \geq 0.88 \wedge \widehat{W}_2^{\text{out}} \leq 4.5$) and 89.0% under the strict 3-way threshold adding $\rho_{\text{grad}} \geq 0.88$, with $\text{SSIM}_{\text{edge}} = 0.9967$, non-circular $\text{SSIM}_{\text{full}} = 0.9752$, $\rho_{\text{grad}} = 0.8943$, and $\widehat{W}_2^{\text{out}} = 1.96$ ($\widehat{SW}_2^{\text{eval}} = 2.07$) in 116.2 ms mean CPU latency ($P_{50} = 93.3$ ms, $P_{95} = 235.0$ ms).
- **Global vs. Spatially Masked ROI Wavelet Locking:** Enabling 5×5 coherence-weighted ROI masking (**Agent-GhibliLock ROI-Masked**) locks active foreground emblems and rails ($\text{SSIM}_{\text{edge}} = 0.9977, \rho_{\text{grad}} = 0.8899$) while leaving non-ROI background regions in smooth Ghibli gouache brushwork, further improving post-export color fidelity to $\widehat{W}_2^{\text{out}} = 1.75$ ($\widehat{SW}_2^{\text{eval}} = 1.87$) and suppressing background high-frequency sensor grain by $> 55\%$ relative to global locking.
- **Haar DWT Depth Ablation ($K \in \{1, 2, 3\}$), Iteration-Proportional Latency, and Stopping Index Distribution:** Table II reports both per-stratum metrics and the multi-level wavelet depth ablation ($n = 60$ trials). Single-pass CPU execution at 64×64 requires $T_{\text{pass}} \approx 88$–$104$ ms, so total closed-loop latency scales linearly with the mean number of controller iterations $\bar{\tau}$ (Latency $\approx \bar{\tau} \times T_{\text{pass}}$): $K = 1$ ($\bar{\tau} = 1.05, 96.4$ ms) achieves the lowest $\widehat{W}_2^{\text{out}} = 1.4160$ but lower full-band $\text{SSIM}_{\text{full}} = 0.8613$; $K = 2$ ($\bar{\tau} = 1.77, 156.2$ ms, our default) balances $\text{SSIM}_{\text{edge}} = 0.9982$, $\text{SSIM}_{\text{full}} = 0.9744$, and $\widehat{W}_2^{\text{out}} = 1.6830$; and $K = 3$ converges in a single pass ($\bar{\tau} = 1.00, 104.4$ ms), maximizing $\text{SSIM}_{\text{full}} = 0.9773$ and $\rho_{\text{grad}} = 0.9724$ at a higher $\widehat{W}_2^{\text{out}} = 2.2725$. Across the $N = 600$ benchmark, the exact closed-loop stopping index distribution is $\tau = 1 : 508/600$ (84.67%), $\tau = 2 : 15/600$ (2.50%), $\tau = 3 : 3/600$ (0.50%), and $\tau = 4 : 74/600$ (12.33%, summing to 100.00%); under the $n = 100$ high-contrast stress suite ($\tau_{\text{struct}} = 0.92, \tau_{\text{color}} = 3.2$), the decoupled $(\tau_\gamma, \tau_\alpha)$ controller converges at $\tau = 1 : 64.00\%, \tau = 2 : 2.00\%, \tau = 3 : 0.00\%, \tau = 4 : 34.00\%$.
- **Video GOP Deflickering (S_4) Across 5 Temporal Baselines:** Across $n = 100$ trials of 64×64, 8-frame GOP sequences, unconstrained synthesis exhibits $\sigma^2_{\text{flicker}} = 347.92$. Unwarped Pixel-Space EMA reduces flicker only to 5.9002 while blurring moving edges ($\text{SSIM}_{\text{edge}} = 0.0870$), and **Motion-Compensated Pixel-Space EMA** (warping each previous RGB frame by $(\Delta u_t, \Delta v_t)$ prior to EMA) still suffers $\sigma^2_{\text{flicker}} = 5.9489$ and low $\text{SSIM}_{\text{edge}} = 0.1614$ because candidate frames lack high-frequency source phase. Global Luminance Histogram Locking eliminates mean flicker ($\sigma^2_{\text{flicker}} = 0.0002$) but fails to restore spatial edges ($\text{SSIM}_{\text{edge}} = 0.0871$). **Ungated Motion-Compensated Wavelet EMA** ($\beta_t \equiv \beta$) achieves $\sigma^2_{\text{flicker}} = 0.0003$ and $\text{SSIM}_{\text{edge}} = 0.9963$ under rigid shifts, but ghosts across non-rigid moving foreground boundaries (where innovation gating improves $\text{SSIM}_{\text{edge}}$ by > 0.50). Agent-GhibliLock's translation-compensated, innovation-gated wavelet EMA achieves $\sigma^2_{\text{flicker,pre-clamp}} < 10^{-12}$, post-sRGB $\sigma^2_{\text{flicker}} = 0.0003$, a 91.0% reduction in local LL_K tile variance ($0.2892 \to 0.0259$), and $\text{SSIM}_{\text{edge}} = 0.9982$.

C. Live Multi-Model Vertex AI Evaluation and Disaggregated Adversarial Provenance (S_5)

Table III reports the live Vertex AI Stage-0 VLM evaluation across `gemini-2.5-flash`, `gemini-2.5-pro`, and `gemini-3-flash-preview` alongside the 5 synthetic drift curriculum tiers (C1–C5). In Part 1, each live VLM inspects the source asset and returns its JSON `structural_retention_prior` $\hat{\eta}_{\text{VLM}} \in [0.05, 0.15]$, which directly sets $\eta_{\text{struct}} = \hat{\eta}_{\text{VLM}}$ in the candidate generator (yielding $\text{SSIM}^{\text{pre}}_{\text{edge}} \in [0.1150, 0.1843]$ prior to Agent-GhibliLock compilation). Note that $\text{PSNR}_{\text{QIM}} \approx 47.09$–$47.42$ dB specifically measures Stage-3 QIM watermark embedding fidelity between the post-Step-B stylized image $\widehat{I}$ and the final watermarked output I_{out}, while local CPU compiler latency (354.7–432.5 ms at 128×128) is reported separately from live Vertex AI network API latency (1696.1–3164.0 ms).

Table IV disaggregates the S_5 ($n = 100$) adversarial provenance results across four balanced, mutually exclusive $n = 25$ regimes ($4 \times 25 = 100$ trials): A_1 ($n = 25$, unstripped 2×2 pixel tamper), A_2 ($n = 25$, pure C2PA `caBX` chunk stripping), A_{3a} ($n = 25$, strip + ±2 LSB noise), and A_{3b} ($n = 25$, strip + actual PIL lossy JPEG $Q = 85$ recompression):

- **Export Self-Noise Accounting:** Prior to any external channel attack, uint8 rounding and sRGB clamping introduce an empirical DCT self-noise of $\text{RMS}(\eta^{\text{export}}) = 0.2854$ ($\sigma^2_{\text{round+clamp}} = 0.0821$ vs. theoretical interior $\sigma^2_{\text{round}} = 1/12 \approx 0.0833$, maximum observed $|\eta^{\text{export}}| = 1.2725 < 3.3486 < \Delta_{\text{QIM}}/4 = 3.5$), yielding a pre-attack single-coefficient error rate of $p_e^{\text{export}} = 0.0000$.

TABLE I

STRATIFIED $N = 600$ BENCHMARK RESULTS ACROSS ALL 9 COMPARATIVE PARADIGMS (`BENCHMARK_RESULTS_N600.JSON`). ALL FOUR WASSERSTEIN-2 CHECKPOINTS ($\widehat{W}_2^{\text{StepA}}, \widehat{W}_2^{\text{StepB}}, \widehat{W}_2^{\text{OUT}}, \widehat{SW}_2^{\text{EVAL}}$) ARE RMS DISTANCES IN CIELAB; **STRUCT / JOINT PASS** REPORTS THE STRUCTURAL PASS RATE ($\text{SSIM}_{\text{EDGE}} \geq 0.88 \wedge \rho_{\text{GRAD}} \geq 0.85$) ALONGSIDE THE STRICT 3-WAY JOINT PASS RATE ($\text{SSIM}_{\text{EDGE}} \geq 0.88 \wedge \rho_{\text{GRAD}} \geq 0.88 \wedge \widehat{W}_2^{\text{OUT}} \leq 4.5$). S_5 **TAMPER DET.** IS NORMALIZED OVER THE $n = 100$ PROVENANCE STRATUM (**B4** DETECTS $25/100 = 25.0\%$ ON S_5, I.E., $25/600 = 4.17\%$ OF TOTAL TRIALS). LATENCY REPORTS MEAN (P_{95}) IN MS.

Paradigm / Baseline	$\text{SSIM}_{\text{edge}} \uparrow$	$\text{SSIM}_{\text{full}} \uparrow$	$\rho_{\text{grad}} \uparrow$	$\widehat{W}_2^{\text{StepA}} \downarrow$	$\widehat{W}_2^{\text{StepB}} \downarrow$	$\widehat{W}_2^{\text{out}} \downarrow$	$\widehat{SW}_2^{\text{eval}} \downarrow$	Struct / Joint Pass	S_5 Tamper Det.	DCT Acc.	GOP $\sigma^2_{\text{flicker}} \downarrow$	Latency (ms) [P_{95}]
B1: Unconstrained Prompt-Only	0.1325 ± 0.0485	0.7820	0.6801	21.99	21.99	21.99 ± 0.39	23.11 ± 0.31	0.0% / 0.0%	0.0%	0.0%	347.92	1419.9 (1493.8)
B2: VLM-Judge Retry (3×)	0.0995 ± 0.0475	0.7720	0.5905	19.44	19.44	19.44 ± 0.46	20.71 ± 0.34	0.0% / 0.0%	0.0%	0.0%	250.50	4377.2 (4594.5)
B3 ($M = 4$): Global SWOT Only	0.0586 ± 0.0240	0.5638	0.2291	4.56	4.56	4.49 ± 0.19	4.91 ± 0.21	0.0% / 0.0%	0.0%	0.0%	0.0002	19.7 (24.2)
B3 ($M = 8$): Global SWOT Only	0.0186 ± 0.0170	0.5839	0.4367	**0.71**	**0.71**	$\mathbf{0.71 \pm 0.02}$	$\mathbf{0.82 \pm 0.02}$	0.0% / 0.0%	0.0%	0.0%	0.0002	30.6 (37.5)
B4: Post-Hoc C2PA Wrapper	0.1325 ± 0.0485	0.7820	0.6801	21.99	21.99	21.99 ± 0.39	23.11 ± 0.31	0.0% / 0.0%	25.0%	0.0%	347.92	**3.2** (**3.5**)
B5: WCT^2 Wavelet Transfer [12]	$\mathbf{0.9992 \pm 0.0002}$	**0.9837**	**0.9825**	10.32	10.32	9.68 ± 0.39	10.36 ± 0.48	100.0% / 0.0%	0.0%	0.0%	0.1800	142.8 (143.2)
B6: Image-Conditioned SDEdit [3]	0.0672 ± 0.0296	0.6347	0.2217	14.17	14.17	14.17 ± 1.87	15.37 ± 1.98	0.0% / 0.0%	0.0%	0.0%	132.21	1390.5 (1452.3)
B7: ControlNet 8×-VAE Adapter [4]	0.2325 ± 0.0683	0.7620	0.8948	0.71	3.20	3.20 ± 0.16	3.51 ± 0.18	0.0% / 0.0%	0.0%	0.0%	100.90	1652.1 (1718.4)
Agent-GhibliLock (Global)	$\mathbf{0.9967 \pm 0.0021}$	**0.9752**	**0.8943**	0.87	2.03	$\mathbf{1.96 \pm 0.33}$	$\mathbf{2.07 \pm 0.32}$	**100.0%** / **89.0%**	**100.0%**	**100.0%**	**0.0003**	116.2 (235.0)
Agent-GhibliLock (ROI-Masked)	$\mathbf{0.9977 \pm 0.0013}$	**0.9119**	**0.8899**	0.85	1.83	$\mathbf{1.75 \pm 0.18}$	$\mathbf{1.87 \pm 0.17}$	**100.0%** / **79.3%**	**100.0%**	**100.0%**	**0.0003**	130.5 (252.9)

TABLE II

PER-STRATUM AGENT-GHIBLILOCK PERFORMANCE ($N = 600$) AND MULTI-LEVEL HAAR DWT DEPTH ABLATION ($K \in \{1, 2, 3\}$) INCLUDING NON-CIRCULAR SPATIAL METRICS AND MEAN CONTROLLER ITERATIONS $\bar{\tau}$.

Stratum / Ablation	n	$\text{SSIM}_{\text{edge}}$	$\text{SSIM}_{\text{full}}$	ρ_{grad}	$\widehat{W}_2^{\text{StepA}}$	$\widehat{W}_2^{\text{StepB}}$	$\widehat{W}_2^{\text{out}}$	$\widehat{SW}_2^{\text{eval}}$	$\bar{\tau}$	Lat. (ms)
S_1: Portrait & Emblem	150	0.9982	0.9756	0.8903	0.8452	1.8173	1.7395	1.8577	1.61	116.6
S_2: Perspective Rails	150	0.9946	0.9755	0.9002	0.9042	2.3382	2.2772	2.3830	1.11	98.8
S_3: Natural Foliage	100	0.9950	0.9732	0.9004	0.8967	2.3170	2.2601	2.3686	1.21	108.5
S_4: Video GOP (8-fr)	100	0.9982	0.9757	0.8907	0.8471	1.8317	1.7516	1.8660	1.58	134.4
S_5: Adv. Provenance	100	0.9982	0.9755	0.8894	0.8466	1.8098	1.7320	1.8491	1.57	131.0
Haar Depth $K = 1$	60	0.9977	0.8613	0.8986	0.8833	1.4797	1.4160	1.5372	1.05	96.4
Haar Depth $K = 2$ (Default)	60	0.9982	0.9744	0.8889	0.8342	1.7522	1.6830	1.8074	1.77	156.2
Haar Depth $K = 3$	60	0.9976	0.9773	0.9724	0.8843	2.3063	2.2725	2.3854	1.00	104.4

- **Comparison Against Blind Cox Spread-Spectrum [20] and Single-Block Blind QIM ($R = 1$) [21]:** Under A_1, **B4** detects $25/25$ (100.0%) via container SHA-256 mismatch (yielding $25/100 = 25.0\%$ across S_5), whereas Single-Block Blind QIM ($R = 1$) [21] achieves 0.0% blind spatial tamper localization because without multi-block repetition ($R \geq 4$), a single block provides no cross-block majority consensus $\hat{\mathbf{w}}$ against which to test local coset residuals. Both informed and *blind majority-voted coset localization* (`expected_bits=None`) in Agent-GhibliLock ($R = 4, 5, 16$) localize the exact tampered 8×8 block $(2, 3)$ on 100.0% (25/25) of trials. Under A_2, **B4** drops to 0.0%, and classical additive Cox spread-spectrum DCT [20] (evaluated in the blind regime without host subtraction) achieves only 73.56% bit accuracy due to Host-Signal Interference, whereas Single-Block Blind QIM ($R = 1$) [21] and Agent-GhibliLock achieve 100.0% (BER = 0). Under A_{3a} (± 2 LSB) and A_{3b} (JPEG $Q = 85$), pre-voting single-coefficient flip rates are $p_e = 0.09\%$ and $p_e = 0.42\%$, which multi-block repetition voting ($R = 4$, $R = 5$ odd voting at 64×64, and $R = 16$ at 128×128) corrects to 100.0% bit accuracy and 100.0% exact 64-bit payload recovery. Under harsher lossy JPEG $Q = 70$ recompression at $\Delta_{\text{QIM}} = 14.0$, mid-band quantization step sizes exceed the $\Delta_{\text{QIM}}/4 = 3.5$ Voronoi margin (51.19% post-vote bit accuracy at $R = 5$), defining the operational compression boundary where larger $\Delta_{\text{QIM}} \geq 22.0$ or lower-frequency DCT modes $\{(1, 2), (2, 1)\}$ are required.

D. Real Natural Photographic Evaluation and Resolution Scaling (64^2 to 512^2)

To validate Agent-GhibliLock on complex natural photographs containing broadband textures, organic lighting gradients, skin tones, and fine metallic/embroidered insignia (beyond synthetic piece-wise smooth test scenes), Table V evaluates `grace_hopper.jpg` (visualized at 128×128 in Fig. 1g–i) across four spatial resolutions (64×64 to 512×512, scaling QIM voting redundancy from $R = 4$ to $R = 256$ votes/bit) and `Minduka_Present_Blue_Pack.png` at 128×128 across all key baselines (**B1**, **B3** $M = 8$, **B5 WCT^2**, **B6 SDEdit**, **B7 ControlNet** 8×**-VAE**, and **Agent-GhibliLock Global / ROI-Masked**). Across all five real-photograph configurations, **B3** ($M = 8$) achieves the lowest $\widehat{W}_2^{\text{out}} = 0.5567$–$0.8408$ but destroys portrait contours ($\text{SSIM}_{\text{edge}} = 0.1894$–$0.2801$), while **B5 (WCT^2)** preserves edges ($\text{SSIM}_{\text{edge}} = 0.9888$–$0.9935$) but leaves large color transport errors ($\widehat{W}_2^{\text{out}} = 4.5573$–$5.4227 > 4.5$). **Agent-GhibliLock (Global)** converges in a single pass ($\bar{\tau} = 1$) across all five configurations, elevating $\text{SSIM}_{\text{edge}}$ from 0.2392–0.3937 (**B1**) and 0.4299–0.5661 (**B7**) to 0.9929–0.9966, achieving non-circular $\text{SSIM}_{\text{full}} = 0.9529$–$0.9712$ and $\rho_{\text{grad}} = 0.9365$–$0.9703$, contracting $\widehat{W}_2^{\text{out}}$ to 1.1108–2.3072, and maintaining BER = 0.0000 under both clean PNG export and lossy JPEG $Q = 85$ recompression. **Agent-GhibliLock (ROI-Masked)** further lowers $\widehat{W}_2^{\text{out}}$ to 0.9906–2.0677 by restricting high-frequency detail locking to high-coherence portrait contours and insignia.

IV. LIMITATIONS, OPERATIONAL SCOPE, AND CONCLUSION

A. Limitations and Operational Failure Modes

Four operational boundary conditions govern production deployments of Agent-GhibliLock:

1) **Upstream Macro-Registration Requirement in Both Global and Foreground ROI Modes:** Because 2D Haar wavelet detail subbands (LH_k, HL_k, HH_k) are spatially localized within $2^k \times 2^k$ dyadic tiles on the source pixel grid I_{src}, detail locking requires that the upstream diffusion candidate I_{cand} be macro-spatially co-registered

TABLE III

PART 1: LIVE GOOGLE CLOUD VERTEX AI STAGE-0 DECOMPOSER EVALUATION (PROJECT="<GCP-PROJECT-ID>", 128×128), WHERE EACH VLM'S EMITTED STRUCTURAL_RETENTION_PRIOR DRIVES CANDIDATE GENERATION. PART 2: CONTROLLED STRUCTURAL DRIFT CURRICULUM TIERS (C1–C5). $\mathrm{PSNR}_{\mathrm{QIM}}$ MEASURES STAGE-3 QIM WATERMARK DISTORTION RELATIVE TO POST-STEP-B $\widehat{I}$.

Evaluation Track	Model / Curriculum Tier	API Lat. (ms)	Tokens (P/C)	$\mathrm{SSIM}_{\mathrm{edge}}^{\mathrm{pre}}$	$\mathrm{SSIM}_{\mathrm{edge}}^{\mathrm{post}}$	$\widehat{\mathbf{W}}_{\mathbf{2}}^{\mathrm{StepA}}$	$\widehat{\mathbf{W}}_{\mathbf{2}}^{\mathrm{out}}$	$\widehat{\mathbf{SW}}_{\mathbf{2}}^{\mathrm{eval}}$	$\mathrm{PSNR}_{\mathrm{QIM}}$	CPU Lat. (ms)
Part 1: Live Vertex AI	gemini-2.5-flash (GA)	1696.09	105/65	0.1150	0.9742	0.7367	1.4421 ± 0.06	1.6016	47.36 dB	354.66
	gemini-2.5-pro (GA)	3163.96	105/11	0.1444	0.9778	0.6877	1.3800 ± 0.06	1.5357	47.39 dB	363.34
	gemini-3-flash-preview	2100.63	105/80	0.1843	0.9802	0.6456	1.3563 ± 0.04	1.5139	47.33 dB	363.96
Part 2: Drift Curriculum	Tier C1 ($\eta_{\mathrm{struct}} = 0.04$)	—	—	0.1199	0.9796	0.7443	1.3926	1.5381	47.42 dB	383.89
	Tier C2 ($\eta_{\mathrm{struct}} = 0.08$)	—	—	0.1981	0.9836	0.6527	1.3435	1.4938	47.39 dB	432.51
	Tier C3 ($\eta_{\mathrm{struct}} = 0.12$)	—	—	0.2707	0.9856	0.5013	1.2514	1.4246	47.09 dB	374.07
	Tier C4 ($\eta_{\mathrm{struct}} = 0.16$)	—	—	0.3418	0.9876	0.4182	1.1877	1.3725	47.14 dB	384.34
	Tier C5 ($\eta_{\mathrm{struct}} = 0.20$)	—	—	0.4325	0.9908	0.4055	1.1536	1.3383	47.34 dB	398.91

TABLE IV

DISAGGREGATED STRATUM S_5 ($n = 100$, FOUR DISJOINT $n = 25$ REGIMES) ADVERSARIAL PROVENANCE MATRIX (BENCHMARK_RESULTS_N600.JSON) COMPARING B4 (C2PA), BLIND COX SS [20], SINGLE-BLOCK BLIND QIM ($R = 1$) [21], AND AGENT-GHIBLILOCK ($R = 4, R = 5, R = 16$).

Adversarial Regime ($S_5, n = 100$)	B4: C2PA	Blind Cox [20]	1-Blk QIM [21]	Pre-Vote p_v	Ghibli ($R = 4$)	Ghibli ($R = 5$)	Ghibli ($R = 16$)
A_1: Unstripped 2×2 Tamper ($n = 25$)	100.0%	0.0%	0.0%	—	100.0%	100.0%	100.0%
A_2: Pure caBX Chunk Strip ($n = 25$)	0.0%	73.56%	100.0%	0.00%	100.0%	100.0%	100.0%
A_{3a}: Strip + ± 2 LSB Noise ($n = 25$)	0.0%	73.12%	100.0%	0.09%	100.0%	100.0%	100.0%
A_{3b}: Strip + Lossy JPEG $Q = 85$ ($n = 25$)	0.0%	78.31%	100.0%	0.42%	100.0%	100.0%	100.0%

with I_{src} (e.g., via image-conditioned SDEdit [3] or ControlNet [4]). While Spatially Masked ROI Wavelet Locking (M_{ROI}) suppresses high-frequency sensor grain and prevents ghosting across *non-ROI background regions* ($M_{\mathrm{ROI}}(x, y) = 0$), within the active *foreground ROI itself* ($M_{\mathrm{ROI}}(x, y) > 0$), Haar detail injection remains anchored to the source pixel grid. Consequently, if an unconditioned generator translates, rotates, or rescales the foreground object pose, locking $D_k(L^*_{\mathrm{src}})$ without prior spatial alignment superimposes source edges onto misaligned candidate chroma, requiring optical-flow or homography pre-warping of I_{cand} to I_{src} prior to Stage 1.

2) **Non-Rigid Motion and Local Parallax in Video GOPs:** In Stage 2's temporal GOP extension (Eq. 17), inter-frame motion compensation estimates a single global integer-tile translation $(\Delta u_t, \Delta v_t)$ per frame on LL_K. When a video scene contains multi-layer depth parallax or rapid non-rigid foreground articulation that deviates from global translation, the local motion innovation $\Delta^{(t)}_{\mathrm{mot}}(u, v)$ exceeds σ_{mot} and drives the gate $\beta_t(u, v) \to 0$. While gating $\beta_t \to 0$ prevents ghosting artifacts, it locally suspends temporal EMA smoothing inside the non-rigid region; achieving full temporal deflickering across deforming objects requires dense optical-flow warping [6] on the wavelet subbands.

3) **Linear $O(HW)$ Filterbank vs. Log-Linear $O(D \cdot HW \log(HW))$ Sorting Latency:** While the 2D Haar DWT and 8×8 block-DCT passes scale linearly in $O(HW)$ operations and Step-A 1D quantile sorting scales log-linearly in $O(D \cdot HW \log(HW))$, pure Python/NumPy execution on CPU takes 90.7/90.1 ms (Global / ROI-Masked) at 64^2, 358.4/357.5 ms at 128^2, 1597.9/1615.8 ms at 256^2, and 6929.9/7895.9 ms at 512^2 on grace_hopper.jpg. Real-time megapixel (1024×1024) compilation therefore requires batched C++/SIMD or CUDA radix-sort and 2D-DCT kernels.

4) **Geometric Desynchronization, Valumetric Scaling, and Symmetric Key Scope in Stage 3:** While our mid-band 8×8 AC DCT QIM watermark achieves 100% bit recovery and blind coset tamper localization under metadata stripping, ± 2 LSB noise, and lossy JPEG $Q = 85$ recompression, three threat vectors bound its scope: (i) non-grid-aligned cropping, aspect-ratio rescaling, or rotation desynchronize the 8×8 block lattice and require template resynchronization or geometric-invariant neural watermarks [10], [19]; (ii) multiplicative valumetric contrast or gamma scaling ($Y \mapsto \gamma_{\mathrm{val}} Y$) scales AC coefficients relative to the fixed step size Δ_{QIM}, requiring pre-decoding block AC RMS gain normalization; and (iii) both the c2sh HMAC-SHA256 digest and keyed dither K_{secret} provide symmetric-key microservice verification, whereas open zero-trust public verification requires asymmetric COSE_Sign1 X.509 signatures [7].

B. Conclusion

We have presented **Agent-GhibliLock**, a deterministic, zero-GPU-dependency post-processing filterbank and provenance compiler that resolves *The Captioning Conundrum* in macro-registered cross-modal VLM-to-Diffusion pipelines. Theoretically, we proved that orthonormal K-level 2D Haar wavelet detail locking preserves exact spatial mean luminance under arbitrary spatial ROI masks $M_k(u, v) \in [0, 1]$ (Theorem 1), established an exact deterministic $(3HW)^{-1/2} \|\widehat{L}^* - \widetilde{L}^*_{\mathrm{cand}}\|_F$ RMS Sliced-Wasserstein perturbation bound across orthogonal frames $Q_m \in O(3)$ (Theorem 2), and derived exact mid-band AC DCT ℓ_1 rounding bounds ($\leq 3.3486 < \Delta_{\mathrm{QIM}}/4 = 3.5$) alongside combinatorial majority-voting conditions for blind block-level coset tamper localization (Theorem 3).

Empirically, across a stratified $N = 600$ benchmark, live multi-model Google Cloud Vertex AI evaluations (gemini-2.5-flash, gemini-2.5-pro, gemini-3-flash-preview), and multi-resolution natural photographs (grace_hopper.jpg at 64^2–512^2 and Minduka_Present_Blue_Pack.png), Agent-GhibliLock simultaneously achieves $\mathrm{SSIM}_{\mathrm{edge}} = 0.9967$–$0.9977$, non-circular $\mathrm{SSIM}_{\mathrm{full}} = 0.9752$, $\rho_{\mathrm{grad}} = 0.8943$, $\widehat{W}_2^{\mathrm{out}} = 1.75$–$1.96$, 99.9% GOP flicker

TABLE V

REAL NATURAL PHOTOGRAPHIC BENCHMARK AND RESOLUTION SCALING (64×64 TO 512×512) ACROSS ALL KEY PARADIGMS ON GRACE_HOPPER.JPG AND MINDUKA_PRESENT_BLUE_PACK.PNG (BENCHMARK_RESULTS_N600.JSON). ALL AGENT-GHIBLILOCK RUNS CONVERGE IN $\bar{\tau} = 1$ ITERATION.

Real Photograph Asset	Method / Paradigm	Votes R	$\text{SSIM}_{\text{edge}} \uparrow$	$\text{SSIM}_{\text{full}} \uparrow$	$\rho_{\text{grad}} \uparrow$	$\widehat{\mathbf{W}}_2^{\text{StepA}} \downarrow$	$\widehat{\mathbf{W}}_2^{\text{out}} \downarrow$	$\widehat{\mathbf{SW}}_2^{\text{eval}} \downarrow$	JPEG85 BER	$\bar{\tau}$	CPU Lat. (ms)
grace_hopper.jpg (64×64)	**B1**: Stylized Candidate	—	0.2392	0.4440	0.5482	2.7708	2.7708	3.0164	—	—	—
	B3 ($M = 8$): Global SWOT Only	—	0.1894	0.3788	0.5123	0.8082	**0.8408**	**1.1167**	—	—	25.3
	B5: WCT^2 Wavelet Transfer [12]	—	0.9935	0.9483	**0.9979**	4.5646	4.5573	4.6753	—	—	142.6
	B6: SDEdit ($s = 0.55$) [3]	—	0.2076	0.4181	0.5393	1.7352	1.7352	1.9347	—	—	—
	B7: ControlNet 8×-VAE [4]	—	0.4936	0.5968	0.7600	0.8082	1.8600	1.9441	—	—	1646.5
	Agent-GhibliLock (Global)	4	**0.9951**	**0.9532**	0.9454	0.8082	2.3072	2.3132	**0.0000**	1	90.7
	Agent-GhibliLock (ROI-Masked)	4	0.9845	0.7986	0.8923	0.8082	2.0677	2.1096	**0.0000**	1	90.1
grace_hopper.jpg (128×128)	**B1**: Stylized Candidate	—	0.3653	0.5984	0.6432	2.8651	2.8651	3.1753	—	—	—
	B3 ($M = 8$): Global SWOT Only	—	0.2796	0.5039	0.5859	0.7208	**0.7894**	**1.0693**	—	—	83.6
	B5: WCT^2 Wavelet Transfer [12]	—	0.9913	0.9563	**0.9964**	4.8213	4.8117	5.0905	—	—	144.8
	B6: SDEdit ($s = 0.55$) [3]	—	0.3145	0.5591	0.6226	1.7803	1.7803	1.9947	—	—	—
	B7: ControlNet 8×-VAE [4]	—	0.5489	0.6912	0.7659	0.7208	1.6743	1.7655	—	—	1706.5
	Agent-GhibliLock (Global)	16	**0.9961**	**0.9604**	0.9403	0.7208	1.5889	1.6969	**0.0000**	1	358.4
	Agent-GhibliLock (ROI-Masked)	16	0.9881	0.8185	0.8893	0.7208	1.4451	1.5824	**0.0000**	1	357.5
grace_hopper.jpg (256×256)	**B1**: Stylized Candidate	—	0.3927	0.6840	0.6519	2.9533	2.9533	3.3096	—	—	—
	B3 ($M = 8$): Global SWOT Only	—	0.2801	0.5760	0.6043	0.6529	**0.7191**	**0.9411**	—	—	296.6
	B5: WCT^2 Wavelet Transfer [12]	—	0.9930	0.9657	**0.9955**	5.0882	5.0794	5.5132	—	—	151.4
	B6: SDEdit ($s = 0.55$) [3]	—	0.3321	0.6394	0.6359	1.8557	1.8557	2.0574	—	—	—
	B7: ControlNet 8×-VAE [4]	—	0.5661	0.7544	0.8099	0.6529	1.6129	1.6836	—	—	1935.4
	Agent-GhibliLock (Global)	64	**0.9966**	**0.9661**	0.9398	0.6529	1.2859	1.3948	**0.0000**	1	1597.9
	Agent-GhibliLock (ROI-Masked)	64	0.9908	0.8287	0.8978	0.6529	1.1854	1.3112	**0.0000**	1	1615.8
grace_hopper.jpg (512×512)	**B1**: Stylized Candidate	—	0.3937	0.6922	0.7201	3.0279	3.0279	3.4062	—	—	—
	B3 ($M = 8$): Global SWOT Only	—	0.2636	0.5837	0.6901	0.6374	**0.6957**	**0.8808**	—	—	1118.5
	B5: WCT^2 Wavelet Transfer [12]	—	0.9912	**0.9714**	**0.9944**	5.2202	5.2111	5.7497	—	—	177.2
	B6: SDEdit ($s = 0.55$) [3]	—	0.3074	0.6459	0.7123	1.8840	1.8840	2.0804	—	—	—
	B7: ControlNet 8×-VAE [4]	—	0.5378	0.7413	0.8375	0.6374	1.4556	1.5298	—	—	3053.2
	Agent-GhibliLock (Global)	256	**0.9944**	0.9712	0.9365	0.6374	1.1108	1.2259	**0.0000**	1	6929.9
	Agent-GhibliLock (ROI-Masked)	256	0.9878	0.7656	0.8851	0.6374	0.9906	1.1188	**0.0000**	1	7895.9
Minduka_Present_Blue_Pack.png (128^2)	**B1**: Stylized Candidate	—	0.2642	0.6507	0.7932	4.1654	4.1654	4.5888	—	—	—
	B3 ($M = 8$): Global SWOT Only	—	0.2720	0.6232	0.7603	0.5575	**0.5567**	**0.7292**	—	—	68.2
	B5: WCT^2 Wavelet Transfer [12]	—	0.9888	0.9511	**0.9978**	5.4729	5.4227	5.6888	—	—	144.3
	B6: SDEdit ($s = 0.55$) [3]	—	0.2552	0.6420	0.7820	2.2131	2.2131	2.4393	—	—	—
	B7: ControlNet 8×-VAE [4]	—	0.4299	0.7500	0.8695	0.5575	1.6176	1.6565	—	—	1698.8
	Agent-GhibliLock (Global)	16	**0.9929**	**0.9529**	0.9703	0.5575	1.6623	1.7677	**0.0000**	1	310.5
	Agent-GhibliLock (ROI-Masked)	16	0.9885	0.8053	0.9118	0.5575	1.5045	1.5942	**0.0000**	1	308.6

suppression, and 100.0% blind watermark bit recovery under container stripping, ± 2 LSB noise, and lossy JPEG $Q = 85$ recompression. Future work will extend Agent-GhibliLock with GPU/SIMD-accelerated radix-sort and DCT kernels for sub-20 ms megapixel compilation, dense optical-flow wavelet warping for non-rigid video GOPs, and hybrid integration of asymmetric COSE_Sign1 C2PA manifests with geometrically invariant watermarks.

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